

WATERSHED  
MANAGEMENT  
MODELING SYSTEM  
VERSION 2.0



# PHASE II REPORT: BMP MODEL AND OPTIMIZATION FRAMEWORK



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# 1 OVERVIEW

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Since its release by the Los Angeles County Flood Control District (LACFCD) and Los Angeles County Department of Public Works (LACDPW) in 2009, the Watershed Management Modeling System (WMMS) has played an important role in supporting watershed planning (LARWQCB 2018), best management practice (BMP) conceptual design and implementation, and climate change analyses for the region (USBR and LACDPW 2016). At the time of its release, WMMS was a state-of-the-art stormwater planning tool based on the best available data and models. Recent advancements in computational efficiency, BMP parameterization, monitoring data, and high-resolution modeling and data have provided an opportunity to enhance and update WMMS. To incorporate these advances and develop an accessible, user-focused web platform, LACFCD and LACDPW conducted an update to WMMS (WMMS 2.0), which was focused into two distinct phases.

The Phase I effort provided updates to the baseline hydrology and water quality model for all coastal watersheds in Los Angeles County (County), based on the Loading Simulation Program C++ (LSPC). This included reconfiguration of hydrologic response units (HRUs), meteorological boundary conditions, hydraulic features that influence stormwater conveyance (e.g., dams, spreading grounds), and a hydrologic and water quality calibration and validation process. Calibration and validation utilized the most recent available data from streamflow gages and water quality stations throughout the model domain. The updated LSPC model provides a robust system for hourly simulation of hydrology and water quality throughout the County watersheds. Complete documentation of the Phase I LSPC model development is provided in the *WMMS 2.0 Phase I Report: Baseline Hydrology and Water Quality Model* (WMMS 2.0 Phase I Report).

Phase II focused on updating components of the System for Urban Stormwater Treatment and Analysis Integration (SUSTAIN) model as well as integrating the model into a broader web-based platform to support BMP implementation planning. Developed by the U.S. Environmental Protection Agency (EPA) Office of Research and Development, SUSTAIN is a decision support system that provides process-based simulation and cost-optimization of BMPs. Updates to SUSTAIN included improvements to the model executable and the representation, parameterization, and cost estimation of BMPs to support modeling of hydraulic and water quality treatment processes and enable cost optimization of alternative BMP implementation scenarios. These updates culminated into a next-generational modeling platform based on SUSTAIN. To enable maximum usability and transparency for SUSTAIN applications in the County, the WMMS 2.0 web-based user platform was developed. This platform includes a file repository where the LSPC and SUSTAIN executables and input data can be downloaded, an interactive map and timeseries viewer to illustrate LSPC model inputs and output, and a suite of utilities to facilitate BMP cost optimizations that can inform watershed planning efforts. In concert, the utilities in the WMMS 2.0 web platform provide substantively improved modeling capabilities and user accessibility, further empowering successful stormwater management for County coastal watersheds included in the model domain (Figure 1-1).

This *WMMS 2.0 Phase II Report* is organized into the following sections:

- ▼ **Section 2: Model Background.** The development of SUSTAIN and its relevance in stormwater management planning in the region.
- ▼ **Section 3: Model Updates.** Improvements to the SUSTAIN model, including enhancements for improved functionality and an updated BMP menu with default design and performance parameters and cost functions.
- ▼ **Section 4: WMMS 2.0 Web-based Modeling Platform.** The WMMS 2.0 web application was developed to provide users with a consistent process for accessing and utilizing the updated models, optimization framework, and associated documentation.



Figure 1-1. WMMS 2.0 model domain and the 2,655 subwatersheds represented in LSPC and SUSTAIN.

## 2 MODEL BACKGROUND

Stormwater runoff from urbanized areas is recognized as contributing to the degradation of water quality in downstream waterbodies (Boulanger and Nikolaidis 2003; Ha and Stenstrom 2003; LARWQCB 2012; LARWQCB and USEPA 2005a, 2005b, 2006, 2011; Stein *et al.* 2007; Vassilios and Hamid 1997; Wong *et al.* 1997). As precipitation accumulates and runs off urban surfaces, it has an opportunity to accumulate pollutants before discharging to receiving waters (Ackerman *et al.* 2005, Ackerman and Weisberg 2006, Brown and Peake 2006, Davis *et al.* 2001). Common urban stormwater pollutants originate from both anthropogenic (e.g., car tires, brake pads) and natural sources (e.g., erosion in headwaters) and can include sediment, heavy metals, oil and grease, nutrients, and toxics (Davis and McCuen 2005). Within the last several decades, there has been a coordinated effort to



improve water quality by managing the harmful effects of impaired stormwater runoff by implementing stormwater control measures such as BMPs (Davis *et al.* 2003, Mandelker 1989, McCuen 2003, USEPA 1999). Hydrologic process-based models such as LSPC and SUSTAIN have been used extensively to evaluate stormwater runoff (Crawford and Burges 1967, USEPA 1997 and 2008) and the potential for BMPs to improve water quality, and support the development of BMP implementation strategies (Ackerman and Stein 2008, USEPA 2008 and 2017). Released by EPA in 2009, SUSTAIN is an open-source, process-based modeling system developed by the EPA Office of Research and Development for simulating watershed hydrology and hydraulics, water quality, and BMP processes at multiple scales (USEPA 2009). SUSTAIN provides users with a platform for evaluating the cumulative impacts of BMP implementation on urban hydrology and water quality across a watershed area (USEPA 2014).

SUSTAIN's algorithms and components are a composite of several other simulation modules and models integrated in a cohesive and powerful framework (Lai *et al.* 2003, 2004, 2005, 2006; Prince George's County 2001; Riverson *et al.* 2004; Rossman 2008). SUSTAIN's key componentry include the following:

- ▼ **Watershed Module.** Simulates watershed hydrology, transport, and water quality processes through a network of conveyances and model nodes (e.g., BMPs). SUSTAIN also provides the option to link to an existing watershed model using unit-area runoff timeseries and associated area distributions as boundary conditions to BMPs (which can be exported from LSPC). This option to link is more commonly used because it decouples watershed simulation from BMP simulation streamlines the optimization process.
- ▼ **BMP Module.** Simulates process-based treatment of water and pollutants with physically represented ponding layers, soil media, infiltration processes, and percolation of stormwater to underlying native soils. BMPs can be individually configured to represent different types (e.g., dry pond, bioretention), dimensions, routing, performance, and cost.
- ▼ **Optimization Module.** Assesses the impact of varying BMP types, sizes, and configurations by iterating through various implementation strategies. This process uses a non-dominated sorting genetic algorithm II (NSGA-II) to evaluate a solution space of potential BMP configurations to identify a cost-optimal set among multiple interacting and competing scales and factors (e.g., cost, configuration, and effectiveness) (Deb *et al.* 2002).

In 2014, SUSTAIN version 1.2 was released. This update included a release of the model source code as well as an optional graphical user interface (coded as extension to ArcGIS 9.3) for processing spatial data into model input files. Since this release, private organizations and public agencies have adopted, incorporated, and customized SUSTAIN, or duplicated processes within SUSTAIN modules, to support watershed management and BMP implementation planning (CSM and USEPA 2019, Innovyze 2015, Wu *et al.* 2019). SUSTAIN has also been used in numerous applications and research efforts (Gao *et al.* 2015, Gallo *et al.* 2018, Gold *et al.* 2015, Lai *et al.* 2007, Mika *et al.* 2017a, Mika *et al.* 2017b, Read *et al.* 2019, Tariq *et al.* 2017, Wolfand *et al.* 2018, Zou *et al.* 2015).

In 2015, as a component of the original WMMS modeling platform (LACDPW 2011a), SUSTAIN was applied across County coastal watersheds in various Enhanced Watershed Management Programs and Watershed Management Programs (E/WMPs). E/WMPs are watershed-scale programs charged with identifying and implementing strategies, control measures, and BMPs to attain water quality targets pursuant to the Los Angeles County Municipal Separate Storm Sewer System (MS4) Permit Order No. R4-2012-0175 (Permit; LARWQCB 2012). As part of the E/WMP process, Reasonable Assurance Analyses (RAAs) were conducted to demonstrate that applicable water quality-based effluent and receiving water limitations are achieved by quantitatively evaluating various management pathways in a peer-reviewed model environment (e.g., SUSTAIN). This measurable and demonstratable approach provides reasonable assurance of attaining management objectives over time



through an adaptive management approach. SUSTAIN was the most widely applied BMP modeling platform used in the RAAs for simulating and evaluating BMPs to improve the existing water quality condition (Ballona Creek WMG 2016, DC WMG 2016, ESGV WMG 2014, LCC WMG 2015, LSGR WMG 2015, LLAR WMG 2015, Malibu Creek WMG 2016, ULAR WMG 2016, USCR WMG 2016, USGR WMG 2016).

## 3 SUSTAIN MODEL UPDATES

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The WMMS 2.0 modeling framework includes enhancements to SUSTAIN 1.2 to provide additional simulation features, improved transparency of modeling processes and assumptions, and increased accessibility for the user community, resulting in a powerful tool for the evaluation of BMP strategies to improve water quality on a watershed-scale. Updates to SUSTAIN leveraged lessons learned from its broad application in the E/WMPs as well as subsequent use supporting BMP conceptual design and implementation planning. The model updates discussed in this section include: (1) model functionality enhancements, (2) improved BMP characterization and parameterization, and (3) updated BMP cost estimation.

### 3.1 Model Functionality Enhancements

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Several modifications that improve the modeling platform and increase its overall utility were incorporated into SUSTAIN based on multiple watershed- and site-scale applications in recent years. These modifications included advancements in computation efficiency, streamlined linkage to LSPC, increased flexibility in how BMPs are modeled, and an update to the overall optimization process. The SUSTAIN model files (executable and .dll) on the WMMS 2.0 repository are required for WMMS 2.0 applications, older SUSTAIN executables will not function properly within the WMMS 2.0 environment. The following summarizes the SUSTAIN model enhancements for WMMS 2.0:

1. **64-bit simulation platform.** The original SUSTAIN C++ program operated in a 32-bit environment and was compiled in Microsoft Visual Studio 6. SUSTAIN has been upgraded to a 64-bit environment in Microsoft Visual Studio 17. This update is Windows 10 compatible, significantly improving the ability of the model to maximize hardware resources and ensures the software can be easily modified and maintained through future versions of Visual Studio.
2. **Underdrain effluent concentration.** To provide additional flexibility for representing water quality improvement from BMPs, an option for a user-specified pollutant effluent concentration (upper limit) for water flowing through a BMP underdrain outlet was added. The International BMP Database and other repositories for BMP performance summaries often report BMP effluent as a pollutant concentration (WERF 2017). The updated SUSTAIN model can directly incorporate these concentrations in the BMP modeling process. When this option is selected, the simulated effluent concentration will always be less than or equal to the user-specified pollutant concentration. This concentration only applies to the *underdrain* outlet effluent. Flow that overflows or bypasses the BMP will be simulated at the ponded-water or influent concentration, as applicable. Users continue to have the option to simulate underdrain pollutant removal as a percent reduction (e.g., filtration or uptake within BMP soil media, prior to discharge through the underdrain).
3. **BMP diversion simulation.** Most large-scale regional BMPs are off-line projects, meaning that diversion structures are implemented to redirect stormwater from an existing conveyance (e.g., storm drain or channel) to the BMP. In SUSTAIN 1.2, this was represented by adding a functional table (F-table), which is a rating curve with multiple outlets representing the diversion and a bypass. In the updated SUSTAIN model, this has been streamlined to include a maximum inflow rate as a property of the BMP itself, effectively representing a diversion

structure that redirects stormwater from the routing network to the off-line BMP. Additionally, the BMP diversion rate has been added as a decision variable in the optimization module, meaning that a diversion flow rate can now be optimized as part of a BMP sizing analysis. The associated diversion cost can also be expressed as a function of the diversion flow rate.

4. **Bypass fraction.** In the original iteration of SUSTAIN, when a BMP was at capacity during a model simulation (i.e., soil porosity is saturated and ponding volume is full), excess inflow and associated pollutants would continue to enter the BMP and its contents would mix before overflowing. Any volume beyond the BMP's capacity at the end of a timestep would overflow to the next downstream node, transporting any associated pollutants at completely mixed concentrations of the ponded water. While this formulation is acceptable for some BMPs like stormwater ponds, it is not representative of other Low Impact Development (LID) or green infrastructure practices (e.g., bioretention with curb-cut inlet structures). For those BMPs, inflow mixing could result in an overestimate of bypass concentration, especially for those designed to capture "first-flush" and bypass excess inflow. SUSTAIN has been updated to include a bypass fraction as an optional configuration parameter for a modeled BMP. That term is a fraction between 0 and 1, where 0 indicates no bypass (all inflow is completely mixed with ponded water) and 1 means only the volume required to fill the BMP enters and the excess completely bypasses. A value of 0.5 allows up to 50% of the inflow to mix as the BMP nears maximum storage capacity—such a value acknowledges some degree of turbulent mixing at the BMP inlet.
5. **Potential evapotranspiration (PET) timeseries from LSPC.** PET in SUSTAIN can be represented in a daily timeseries or as constant monthly average values. To build a direct linkage between SUSTAIN and LSPC, a new option accepts the hourly PET timeseries from LSPC with diurnal variation as a direct input in lieu of the constant, daily, or monthly PET values.
6. **Fixed effluent discharge rate.** Some regional BMP types are configured to treat stormwater at a maximum treated outflow rate. Examples of this include BMPs that contain cartridge filtration units that discharge treated stormwater back to a waterbody. SUSTAIN has been modified to allow users to enter a maximum discharge rate in cubic feet per second (cfs) for a BMP (typically < 5 cfs). Pollutant effluent concentration using this option is held constant as a user-specified value for each pollutant; however, the modeled concentration is used during the simulation whenever it is lower than the specified effluent limit.
7. **Two-tiered optimization.** While a two-tiered optimization was designed as part of SUSTAIN 1.2, the setup of this process was seldom used and did not have a graphical user interface for convenient configuration. Two-tiered optimization facilitates watershed-scale management planning by layering the problem in stages. First, the watershed is discretized into smaller units (subwatersheds) that are individually optimized in parallel (i.e., Tier 1). Next, the cost-effectiveness curves from the Tier 1 optimization become the inputs to Tier 2, which is evaluated at a point downstream of multiple subwatersheds. During Tier 2, the optimization only needs to search through the set of cost-effective solutions from Tier 1 as decision variables rather than simulating all possible (i.e., thousands) solutions. Additional details on this process are provided in Section 4.3 Optimization Framework.
8. **Stepwise BMP iterator.** A stepwise BMP size routine has been added to SUSTAIN. This option progressively iterates through a single BMP and decision variable (e.g., BMP length) from a minimum value to a maximum value at a consistent step interval. For example, a BMP with a minimum length of 0 feet, maximum length of 100 feet, and a step of 10 feet can be specified. This simulation would step from 0 to 100 in increments of 10 to generate an 11-point cost-effectiveness curve for a single BMP and assessing its complete range of performance.

9. **Direct Output Linkage to LSPC.** LSPC input Card 435 provides a mechanism for linking external timeseries as an input to the watershed model. This linkage file is a convenient format to link timeseries output from SUSTAIN to the LSPC model. This update makes it possible to turn off the simulation of surface runoff and pollutant loading for selected subwatersheds within LSPC and replace them with alternative runoff timeseries results from a subwatershed outlet in SUSTAIN that reflect upstream BMPs in the subwatershed. An output feature has been added to SUSTAIN which will automatically generate the timeseries files pre-formatted for the LSPC linkage feature.

These modifications result in increased computing capability and functionality of SUSTAIN and provide an elevated platform for modeling to create watershed-scale strategies.

## 3.2 BMP Menu and Design Parameters

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Appropriately representing BMPs in SUSTAIN (i.e., parameterization) to determine their corresponding performance relies on several considerations, including design type, dimensions, evapotranspiration processes, and treatment/filtration processes, among others. To represent the various applications and circumstances for BMP planning and design in the region, a variety of BMP configurations were developed for WMMS 2.0. These additions are not inherent to the SUSTAIN model itself, but instead provide a menu of pre-configured BMP types that can be used as a starting point for using WMMS 2.0 to support watershed management planning in the County. Model users have the option of utilizing this menu of BMPs, changing preconfigured “default” parameters, or adding new BMPs not included in this menu.

BMPs and design types in this menu are consistent with those considered in E/WMPs, while also incorporating new and alternative design options as well as applications that meet regional programmatic efforts (e.g., Safe Clean Water [SCW] Program). The WMMS 2.0 Menu of BMPs is presented in Table 3-1. BMPs included in the menu are categorized into two major types: distributed and regional. Distributed BMPs are generally implemented at the street or parcel scale, have small footprints (e.g., street right-of-way), and capture and treat runoff from correspondingly small drainage areas. Regional BMPs are large-scale projects that can treat runoff from larger drainage areas (e.g., hundreds or thousands of acres).

Appropriately and accurately simulating this menu of BMPs in the SUSTAIN model environment relies on the determination of a set of design parameters (e.g., design type, dimensions, and treatment/filtration processes). Descriptions and factsheets for the distributed and regional BMPs included in the menu are presented in Sections 3.2.1 and 3.2.2, respectively. The factsheets list each BMP’s default parameter values for SUSTAIN and the sources for these assumptions. Because specific considerations for BMP design may vary from site to site, the WMMS 2.0 Optimization Framework (Section 4.3) provides users with the option to modify these default parameters during model application.



Table 3-1 WMMS 2.0 Menu of BMPs listed by design type, typical treatment area, and program

| Design Type |                                         | Typical Management Area                                                  | Typical Programmatic Application                                           |
|-------------|-----------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Distributed | Bioretention/biofiltration              | Residential, Public, Commercial, and Industrial Parcels<br>Rights-of-way | Homeowner Incentives, LID Ordinance, LID Retrofits, Green street retrofits |
|             | Porous Pavement                         | Rights-of-way                                                            | Green street retrofits                                                     |
|             | Cistern/Rain Barrel                     | Residential, Public, Commercial, and Industrial Parcels                  | Homeowner Incentives, LID Ordinance, LID Retrofits                         |
|             | Infiltration Well                       | Rights-of-way                                                            | Green street retrofits                                                     |
|             | Diversion to Sewer                      | Various                                                                  | Various                                                                    |
|             | Treatment Unit <sup>1</sup>             |                                                                          |                                                                            |
| Regional    | Infiltration Facility                   | Large Upstream Areas (e.g., neighborhood-scale)                          | Regional BMP Retrofits                                                     |
|             | Treatment Facility, Filtration          |                                                                          |                                                                            |
|             | Treatment Facility, Sewer Diversion     |                                                                          |                                                                            |
|             | Treatment Facility, Retention/Detention |                                                                          |                                                                            |
|             | Treatment Facility, Constructed Wetland |                                                                          |                                                                            |

1. Miscellaneous category of devices that do not conform to the other items listed in the Menu. Can represent stand-alone units (e.g., proprietary tree well) or units in conjunction with a regional project as a pretreatment unit.

### 3.2.1 Distributed BMPs

Distributed BMPs (e.g., LID) provide water quality improvement through several mechanisms, including runoff volume reduction, settling of suspended sediment (and associated pollutants), filtration, and other treatment processes (CASQA 2003, LACDPW 2014, WERF 2017). These BMPs often provide other additional important benefits, including (but not limited to) flood management, traffic calming, neighborhood greening, and reduced heat island effect (USEPA 2015). These types of BMPs generally treat stormwater from small-scale localized areas, and when distributed throughout a watershed, they can have significant cumulative benefits. The following describe the default modeling assumptions of each distributed BMP from the menu and their design variations (e.g., with or without an underdrain).

## Bioretention/Biofiltration

Bioretention practices are designed to mimic the natural hydrologic processes of pre-developed land (CASQA 2003). Broadly, a bioretention unit is a vegetated shallow depressed area supported by soil media to treat stormwater runoff through detention, evapotranspiration, pollutant uptake, filtration through soil media, and/or percolation into native soils. In areas where percolation into native soils is not feasible due to low infiltration rates, an underdrain layer can be implemented to direct treated stormwater to the storm drain network. When implemented with an underdrain layer this configuration is regarded as a *biofiltration* unit. Figure 3-1 shows a xeriscape-style biofiltration unit with drought tolerant vegetation.



Figure 3-1. Drought-tolerant bioretention unit in Southern California.

Principle components of bioretention/biofiltration design include:

- ▼ The depressed nature of bioretention units creates a **ponding area** for stormwater to accumulate and ultimately percolate through the soil media layer.
- ▼ Native **vegetation** provides uptake of stormwater, reduced heat island, micro-scale urban habitat, and neighborhood greening.
- ▼ A **soil media layer** supports vegetation, provides filtration, and storage of stormwater in pore space.
- ▼ In cases where native soils are not conducive to infiltration, an **underdrain layer** is implemented below the media layer. The underdrain layer is gravel-filled and collects and conveys stormwater to the BMP outlet.

Figure 3-2 lists the default model parameters for representing bioretention/ biofiltration in SUSTAIN, including literature and other sources referenced for each assumed parameter.



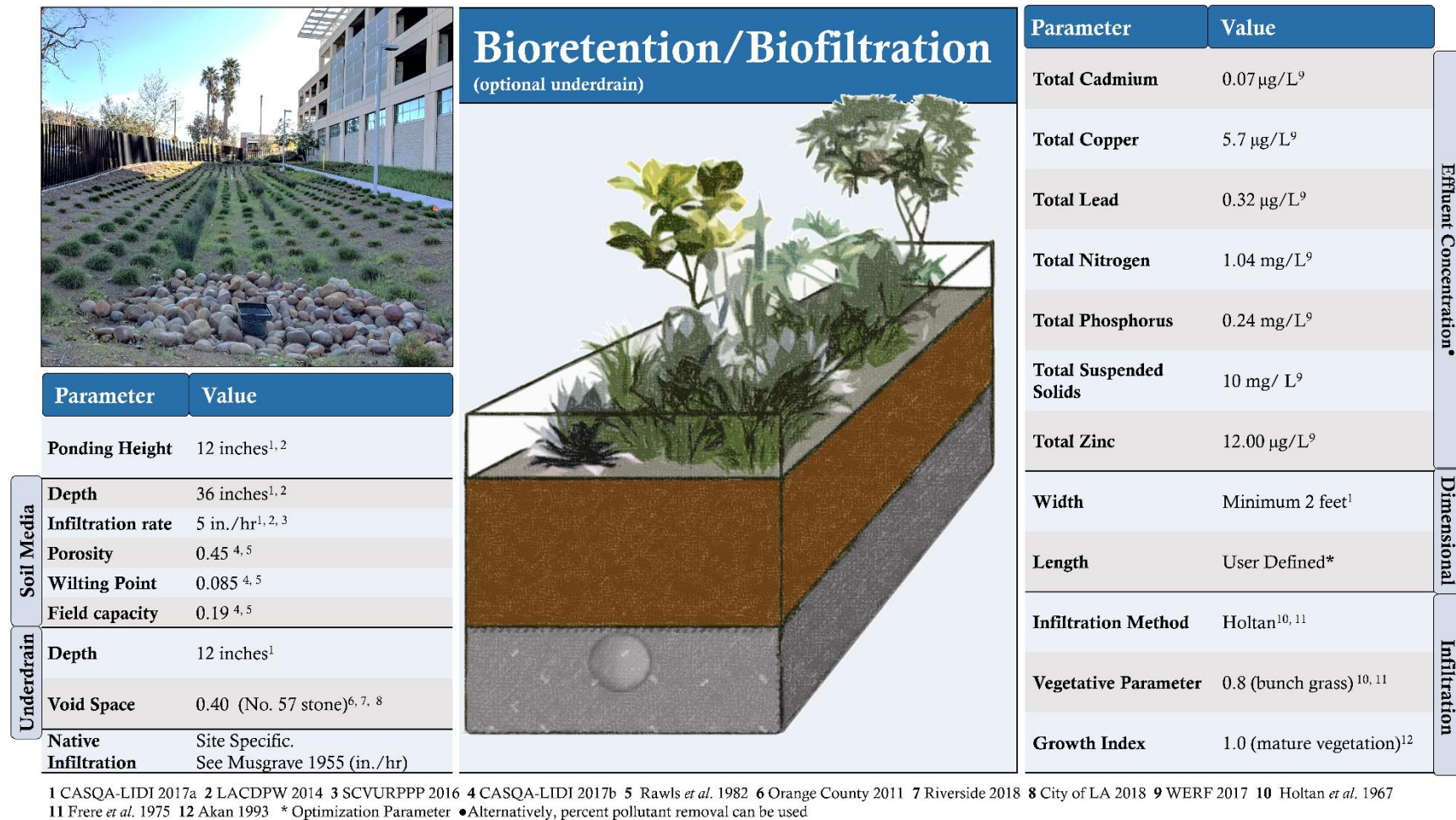


Figure 3-2. BMP parameter factsheet for bioretention/biofiltration.



## Porous Pavement

Porous pavement is a type of roadway or sidewalk material with sufficient pore space for stormwater to be transmitted to subsurface layers (CASQA 2003). A layer of gravel underlays the layer of porous pavement and provides the primary storage component for captured runoff while also providing structural support for the designed traffic/pedestrian load. Porous pavement can most easily be implemented in areas with lower traffic speeds and volumes (e.g., parking lanes, parking lots). In the Los Angeles region, porous pavement is often implemented in conjunction with a bioretention/biofiltration unit for green street designs (City of LA 2009). In areas where infiltration to native soils is not feasible, an underdrain layer can be implemented to direct treated stormwater to the storm drain network.



Figure 3-3. Porous pavement integrated into an LID landscape in Southern California.

Principle components of porous pavement design include:

- ▼ Depending on the application, several **porous material layers** are available for implementation. Porous concrete, asphalt, or gridded pavers are effective options for providing a structurally sound driving or walking platform but also for providing transmissivity of stormwater from the surface to the gravel reservoir layer underneath.
- ▼ The **reservoir layer** is the primary storage component of the porous pavement system. Stormwater is retained before infiltrating to native soils or discharging through the underdrain.
- ▼ Stormwater from the gravel layer will percolate into native soils if conducive to infiltration or, in cases where it is not, an **underdrain layer** is installed to convey treated stormwater. The underdrain is a gravel filled layer that collects and conveys stormwater to the BMP outlet.

Figure 3-4 lists the default parameters for representing porous pavement in SUSTAIN, with and without an underdrain, including literature and other sources referenced for each assumed parameter.



<sup>2</sup> LACDPW 2014 <sup>6</sup> Orange County 2011 <sup>7</sup> Riverside 2018 <sup>8</sup> City of LA 2018 <sup>9</sup> WERF 2017 <sup>10</sup> Holtan *et al.* 1967 <sup>11</sup> Frere *et al.* 1975 <sup>13</sup> City of LA 2014c <sup>16</sup> Collins *et al.* 2010

\* Optimization Parameter • Alternatively, percent pollutant removal can be used

Figure 3-4. BMP parameter factsheet for porous pavement.



## Infiltration Well

Infiltration wells are deep, tubular subsurface units that provide storage of stormwater prior to infiltration. Because of their small surface footprint, infiltration wells can be pragmatic stormwater capture devices in highly developed urban areas. Principle components of infiltration wells include:

- ▼ Infiltration wells provide direct percolation from stormwater runoff into the surrounding native soils, therefore a **pretreatment unit** that captures sediment and other debris is typically implemented in tandem to preserve long-term infiltration integrity of the native soils.
- ▼ The **storage layer** is the portion of the unit that provides the primary stormwater storage component of the unit. This volume is created from the concrete sleeve that is placed below the surface footprint. The design depth of an infiltration well is highly dependent on geologic conditions (e.g., depth to groundwater, stratigraphy). Infiltration wells can be as shallow as 20 feet or as deep as 50 to over 100 feet.
- ▼ The **gravel layer** provides additional storage and support for the unit and is typically implemented as a sleeve around (1-foot thickness) and below (2-foot depth) the storage layer.

Exfiltration from these units is conventionally characterized as a flow rate (cfs) into the subsurface rather than an infiltration rate (in/hr) because of the challenges associated with determining a single representative infiltration rate along the entire depth. Due to the vertical configuration, infiltration is highly dependent on the weight of stored water in the column and varies as the unit draws down. The right side of Figure 3-5 provides a conceptual illustration of an infiltration well (not including the pretreatment unit). The left side of Figure 3-5 shows example tubular concrete units prior to implementation and the default parameters for SUSTAIN.

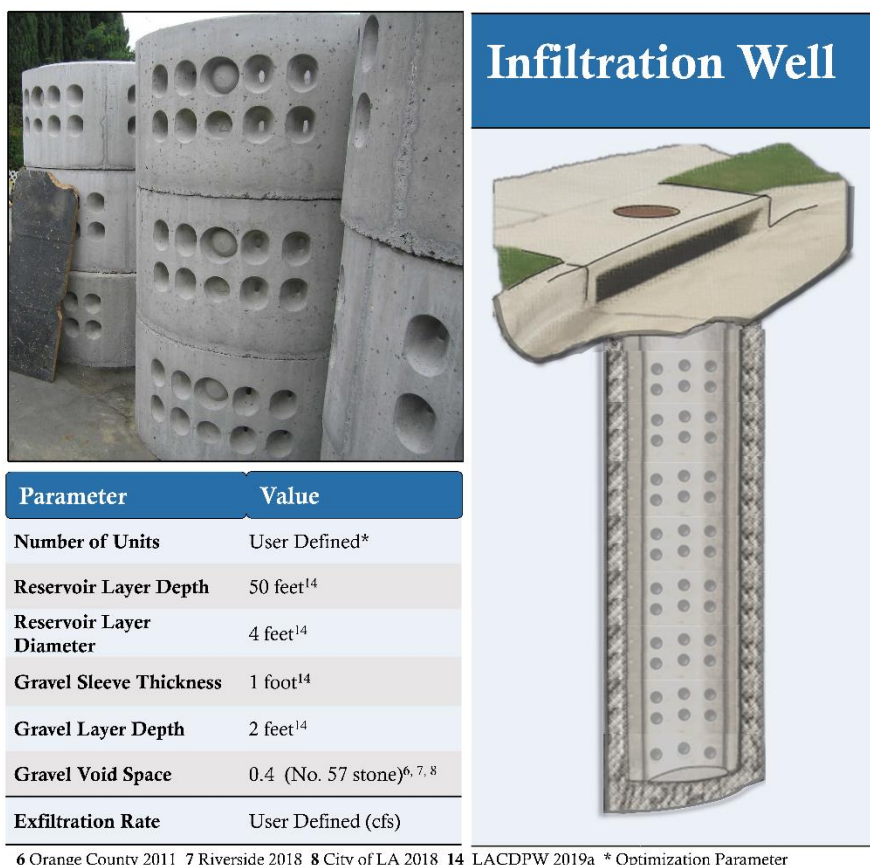


Figure 3-5. BMP parameter factsheet for infiltration wells.



### Cistern/Rain barrel

Rainwater harvesting is the practice of intercepting rainfall from impervious surfaces and storing for beneficial uses or for later release (CASQA 2003). The most common application of rainwater harvesting are cisterns or rain barrels that collect rooftop runoff for outdoor water uses (e.g., watering vegetation). Cisterns are large (500 to over 2,000 gallons) surface or subsurface storage containers, sited at downspouts of commercial, industrial, or multifamily residential properties to collect rainwater from roofs (Figure 3-6). Rain barrels are smaller options more commonly suitable on single family residential properties. In the WMMS 2.0 Menu, cistern parameters (e.g., volume, discharge rate) are simulated as specified by a user, therefore a factsheet with default BMP parameters is not provided.



Figure 3-6. Example cistern located at a roof's downspout.

### Sanitary Sewer Diversion

Diversions to the sanitary sewer can be implemented as large regional projects (described in the following section) or as distributed projects. Distributed diversions to the sanitary sewer generally include a pretreatment unit to remove particulate matter (e.g., trash, debris, sediment), a storage component (Figure 3-7), and an outlet apparatus designed to control outflow at a predetermined rate to the sewer and ultimately to a downstream water reclamation plant (WRP) for treatment. In the WMMS 2.0 Menu, sanitary sewer diversion parameters (e.g., volume, discharge rate) are simulated as specified by a user, therefore a factsheet with default BMP parameters is not provided.



Figure 3-7. Example component of a small-scale sanitary sewer diversion.

### Treatment Unit

Treatment units are a broad categorization of BMPs that include any stormwater device that does not fit the menu of distributed options. For example, they can be small-scale applications that treat parking lot runoff (e.g., filtration, Figure 3-8), pretreatment units (e.g., hydrodynamic separators), or other BMPs whose performance values can be individually parameterized by a user. In the WMMS 2.0 Menu, treatment unit parameters (e.g., volume, discharge rate) are simulated as specified by a user, therefore a BMP factsheet with default parameters is not provided.

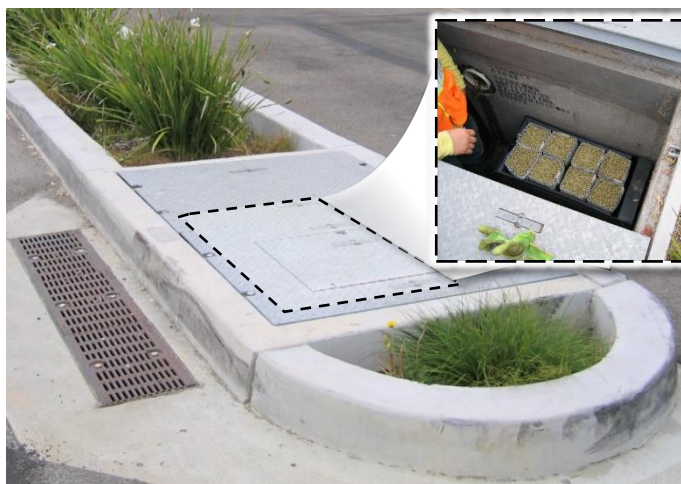


Figure 3-8. Example parking lot treatment unit.

### 3.2.2 Regional BMPs

Like distributed BMPs, regional stormwater capture projects provide water quality improvement in addition to other potential benefits, including (but not limited to) flood management, community enhancement, and/or water supply augmentation (i.e., infiltration to groundwater). Regional facilities have the flexibility to be designed as either surface (e.g., dry detention pond in conjunction with existing open space) or subsurface BMPs for integration into the existing urban landscape (e.g., infiltration galleries below parking lots, parks, or other recreational facilities).

Regional projects can provide water quality improvement of stormwater runoff through detention, infiltration, filtration, and/or water reuse (e.g., on-site irrigation). Generally, these facilities capture stormwater from adjacent channels or storm drains (requires a diversion structure to convey stormwater to the BMP), resulting in a larger capture area and volume as compared to distributed BMPs. Pretreatment units are often implemented with these structures, which capture trash and other debris that can foul the infiltrative capacity of the native soil. Figure 3-9 illustrates a mid-construction photograph of a subsurface infiltration project in the City of Lakewood, CA. This project has an especially large capacity of approximately 3 million gallons.



**Figure 3-9. Subsurface regional BMP in Southern California.**

Regional BMPs are categorized in the WMMS 2.0 Menu as either infiltration facilities or a broad category of treatment facilities which include sanitary sewer diversion units, filtration units, retention/detention, or constructed wetlands. The following summarizes the features and default modeling parameters for each regional BMP category included in the WMMS 2.0 Menu.



## Infiltration Facility

Infiltration facilities are ideal in areas of infiltrative soils to achieve a 100% reduction of captured stormwater volume and associated pollutants through percolation into native soils. When facilities are implemented atop shallow unconfined aquifers, there is an increased opportunity for recharge of local groundwater supplies. Principle components of infiltration facility design include:

- ▼ **Pretreatment and diversion structure.** Diversion structures and pretreatment units are implemented to convey and pretreat stormwater.
- ▼ **Storage layer.** The storage layer is a large void space that retains stormwater before infiltration to the underlying gravel layer. Generally, applications utilize an engineered structure that provides void space for storage. The structure can consist of a range of prefabricated modular units or concrete structures cast on-site. The typical height for subsurface infiltration facilities is between 3 and 12 feet depending on the application.
- ▼ **Gravel layer.** The gravel layer provides support and additional storage underneath the storage layer before infiltrating to native soils.

The left side of Figure 3-10 shows a mid-construction subsurface infiltration facility storage layer and lists the default parameters for SUSTAIN. The right side of Figure 3-10 provides a conceptual illustration of the unit-model storage layer.



| Parameter                | Value                                    |
|--------------------------|------------------------------------------|
| Diversion Rate           | User Defined (cfs)                       |
| Internal Storage Depth   | 10 feet <sup>15</sup>                    |
| Gravel Bedding Depth     | 6 inches <sup>15</sup>                   |
| Gravel Void Space        | 0.40 (No. 57 stone) <sup>6, 7, 8</sup>   |
| Width                    | User Defined (ft)                        |
| Length                   | User Defined* (ft)                       |
| Native Soil Infiltration | Site Specific. See Musgrave 1955 (in/hr) |



6 Orange County 2011 7 Riverside 2018 8 City of LA 2018 15 StormTrap 2019 \* Optimization Parameter

**Figure 3-10. BMP parameter factsheet for subsurface infiltration facilities.**



## Treatment Facility

In areas where infiltration rates are not high enough to efficiently infiltrate stormwater, surface and subsurface treatment facilities provide an alternative mechanism by diverting stormwater to several different potential treatment pathways. Principle components of treatment facility design include:

- ▼ **Pretreatment & diversion structure.** Like infiltration facilities, diversion structures and pretreatment units are implemented to convey and pretreat stormwater.
- ▼ **Storage layer.** Like infiltration facilities, the storage layer is a large void space area that detains stormwater before treatment. Subsurface applications can utilize an engineered structure that increases the available void space for storage. The engineered structure can consist of a range of prefabricated modular structures or concrete structures cast on-site.
- ▼ **Treatment option.** A variety of treatment alternatives are available for implementation of regional treatment facilities. In some cases, multiple treatment pathways can be implemented in one facility to further improve performance.
  - **Sewer Diversion.** Stormwater can be diverted and pumped from the storage layer to a sanitary sewer main for treatment at a downstream WRP. The left side of Figure 3-11 shows a mid-construction subsurface sewer diversion storage layer and lists the default parameters for SUSTAIN. The right side of Figure 3-11 provides a conceptual illustration of the unit-model storage layer.



| Parameter              | Value                 |
|------------------------|-----------------------|
| Diversion Rate         | User Defined (cfs)    |
| Internal Storage Depth | 10 feet <sup>15</sup> |
| Width                  | User Defined (ft)     |
| Length                 | User Defined* (ft)    |
| Discharge Rate         | User Defined (cfs)    |

<sup>15</sup> StormTrap 2019 \*Optimization Parameter

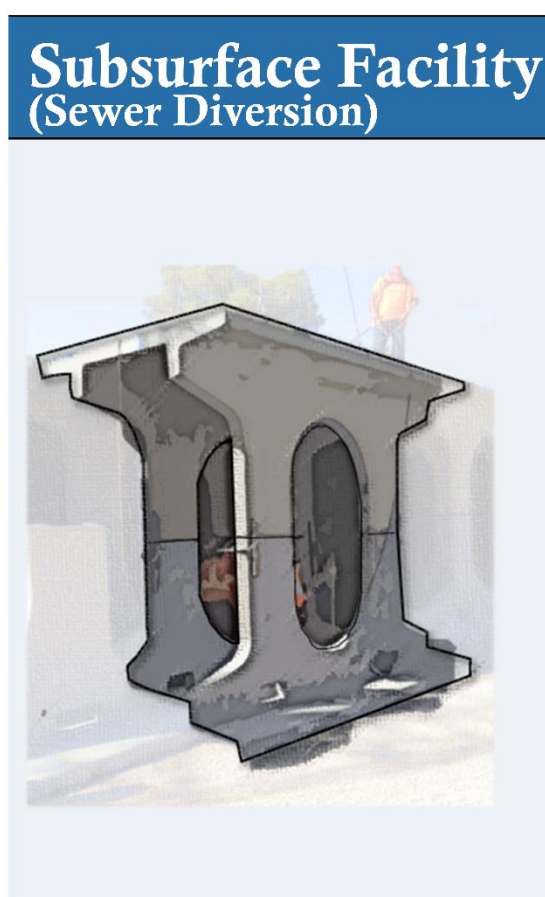
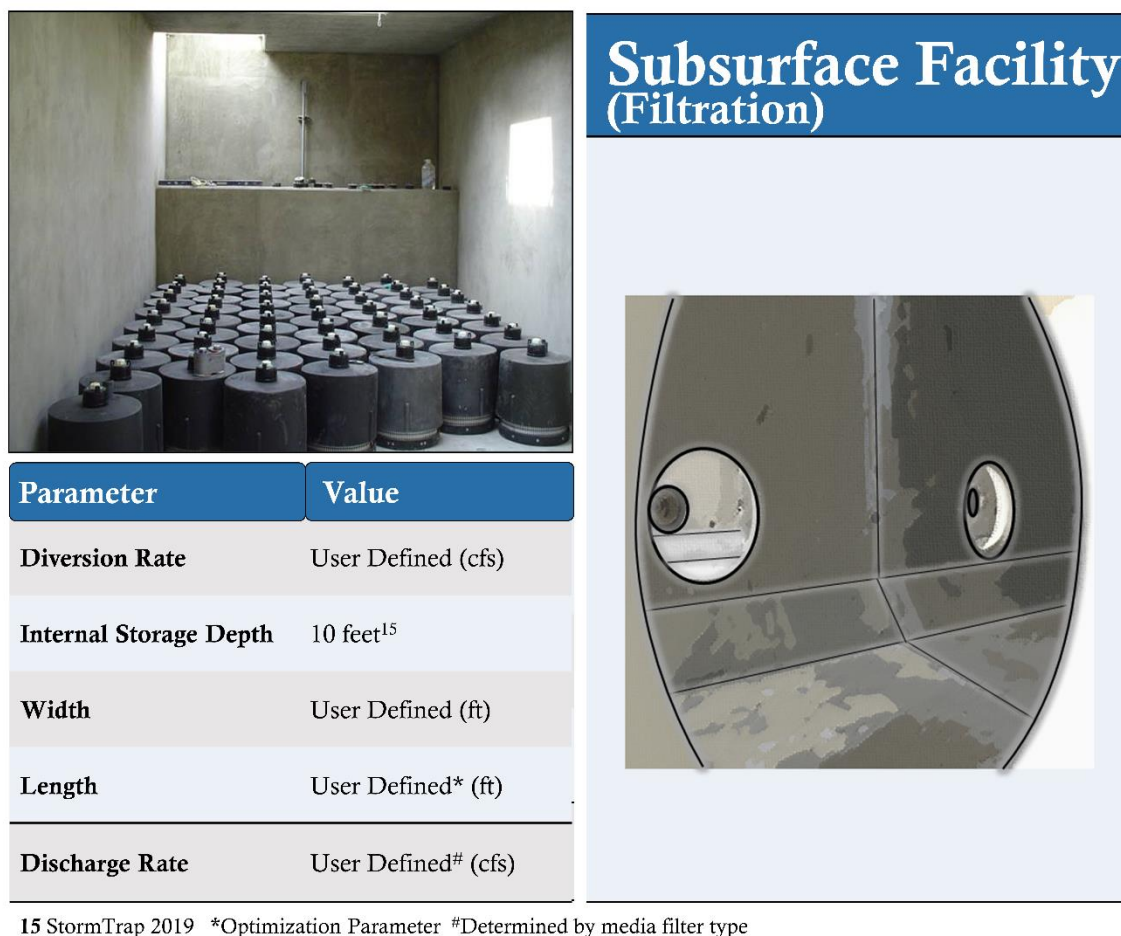


Figure 3-11. BMP parameter factsheet for regional subsurface treatment facilities (sewer diversion).

- **Filter.** A variety of proprietary filtration units are available for implementation at subsurface regional filtration facilities. A regional filtration facility directs stormwater from the storage layer through one or several cartridge-style (or loose media) units. Treated stormwater is discharged back to a proximal storm drain to be conveyed to a downstream receiving water. This style of treatment generally requires a pump to lift diverted stormwater to the inlet of the storage layer or at the BMP outlet to direct treated effluent back to the storm drain network. Pollutant removal performance varies depending on the device selected for implementation. SUSTAIN parameters for constituents of interest should be researched by the user for input into the model. The left side of Figure 3-12 shows a series of cartridge-style filters and lists the default parameters for SUSTAIN. The right side of Figure 3-12 provides a conceptual illustration of the unit-model storage layer.



**Figure 3-12. BMP parameter factsheet for regional subsurface filtration facilities.**

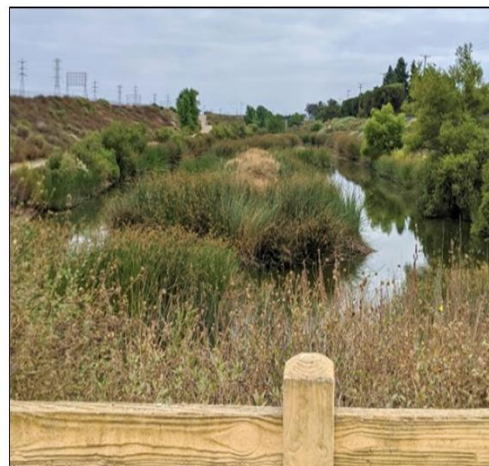
- **Retention and Detention.** Surface retention and detention units create capacity for stormwater in an open surface depression. If soil infiltration rates allow, stormwater is infiltrated into the subsurface (retention) and/or discharged through a riser outlet structure back to the storm drain network (detention). The left side of Figure 3-13 shows a detention unit and lists the default parameters for SUSTAIN. The right side of Figure 3-13 provides default EMC values and a conceptual illustration of the unit-model storage layer.



**Figure 3-13. BMP parameter factsheet for surface detention/retention facilities.**

- **Constructed wetland.** Constructed wetlands emulate natural wetlands with dense vegetation, a sinusoidal flow path, and pools of varying depth. Figure 3-14 lists the default BMP parameters in SUSTAIN and a conceptual illustration of the unit-model storage layer.





| Parameter              | Value                                        |
|------------------------|----------------------------------------------|
| Average Ponding Height | 48 inches <sup>2</sup>                       |
| Soil Media Depth       | 24 inches <sup>2</sup>                       |
| Soil Media Porosity    | 0.45 <sup>4, 5</sup>                         |
| Diversion Rate         | User Defined (cfs)                           |
| Discharge Rate         | User Defined (cfs)                           |
| Native Infiltration    | Site Specific.<br>See Musgrave 1955 (in./hr) |

## Surface Facility (Constructed Wetland)



| Parameter              | Value                                 |                         |
|------------------------|---------------------------------------|-------------------------|
| Total Cadmium          | 0.20 µg/L <sup>9</sup>                | Effluent Concentration* |
| Total Copper           | 6.18 µg/L <sup>9</sup>                |                         |
| Total Lead             | 3.00 µg/L <sup>9</sup>                |                         |
| Total Nitrogen         | 1.43 mg/L <sup>9</sup>                |                         |
| Total Phosphorus       | 0.15 mg/L <sup>9</sup>                |                         |
| Total Suspended Solids | 17 mg/L <sup>9</sup>                  |                         |
| Total Zinc             | 20.00 µg/L <sup>9</sup>               | Dimensional             |
| Width                  | User Defined                          |                         |
| Length                 | User Defined*                         |                         |
| Infiltration Method    | Holtan <sup>10, 11</sup>              | Infiltration            |
| Vegetative Parameter   | 0.8 (bunch grass) <sup>10, 11</sup>   |                         |
| Growth Index           | 1.0 (mature vegetation) <sup>12</sup> |                         |

<sup>2</sup> LACDPW 2014 <sup>4</sup> CASQA-LIDI 2017b <sup>5</sup> Rawls *et al.* 1982 <sup>9</sup> WERF 2017 <sup>10</sup> Holtan *et al.* 1967 <sup>11</sup> Frere *et al.* 1975 <sup>12</sup> Akan 1993

\* Optimization Parameter \*Alternatively, percent pollutant removal can be used

**Figure 3-14. BMP parameter factsheet for surface wetland facilities.**

### 3.3 Default BMP Cost Assumptions

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BMP life-cycle costs were compiled from a variety of local and national data sources including peer-reviewed journal articles, municipal price lists, invoices from completed projects, concept reports, and direct communications with vendors. The resulting default cost functions serve as a foundation for relative comparisons of the cost of BMP implementation for watershed-scale planning across the region. Like the BMP parameters listed above, these default cost assumptions can be modified by the model user to incorporate site-specific considerations, as needed.

For each distributed and regional BMP type in the WMMS 2.0 Menu (Section 3.2), capital and annual operations and maintenance (O&M) costs were developed to formulate BMP life-cycle cost functions. Standard components of capital costs were based on construction costs plus markup costs (contingency, planning, engineering, and design). As determined from several region-specific projects, markup costs were typically 40% of total construction costs (City of LA 2006e, 2006f, 2011, 2012, 2014a, 2014b, 2016a, 2016b, 2017g, 2017i, 2017l). This 40% markup is reflected in the capital costs for all projects below.

#### 3.3.1 Distributed BMP Costs

The following discussions summarize the capital and O&M cost assumptions and associated data sources that informed the development of the default cost functions representing distributed BMPs in SUSTAIN.

##### 3.3.1.1 Capital Costs

Figure 3-15 shows capital costs for bioretention and porous pavement in a box and whisker plot summarizing the range (minimum and maximum, 25<sup>th</sup> and 75<sup>th</sup> percentile), mean (denoted with “x”), and median (center bar). Considered projects were generally provided as a total cost (e.g., bioretention unit: \$250,000). To generate per unit area costs (\$207 per sq. ft. of bioretention), the default BMP parameters from the WMMS 2.0 Menu were applied (see Section 3.2). For cisterns and rain barrels, Figure 3-16 provides a summary of the range, mean, and median capital costs for 178 cisterns, including different costs for the three major material types (stainless steel, galvanized steel, plastic; E-commerce Retailer 2019). Median values for BMPs summarized in the Figure 3-15 and Figure 3-16 and associated data sources are listed in Table 3-2. Capital costs of underdrains for biofiltration and porous pavement units were determined by summing the costs of excavation, underdrain pipe, gravel, and lining garnered from municipal price lists.

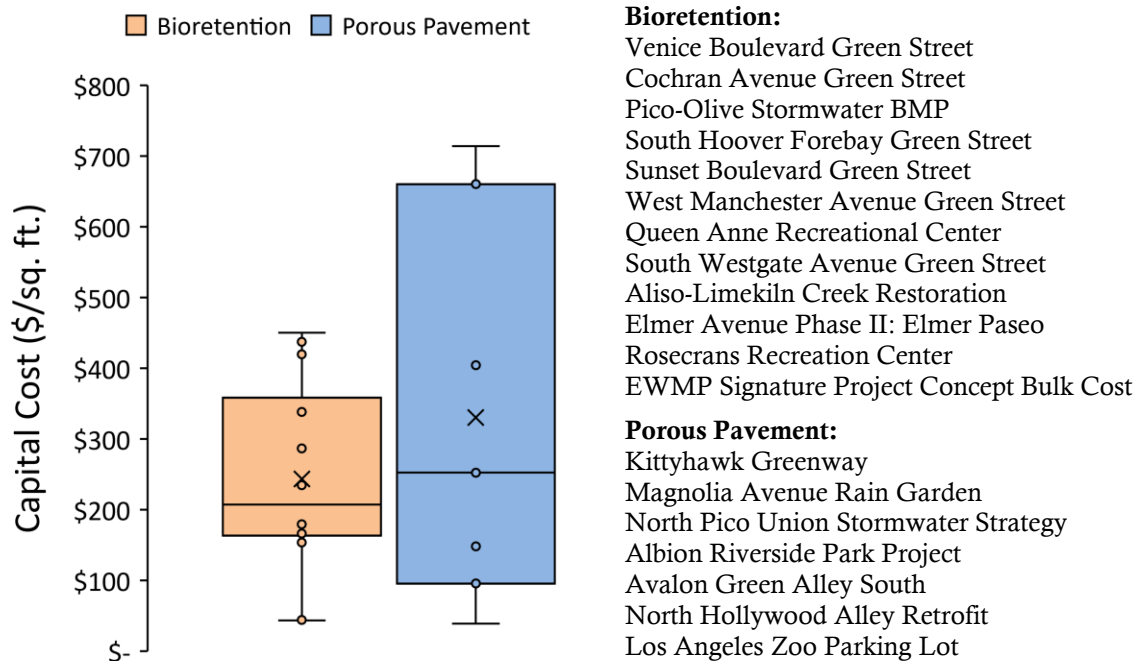


Figure 3-15. Summary of bioretention and porous pavement capital costs and referenced projects.

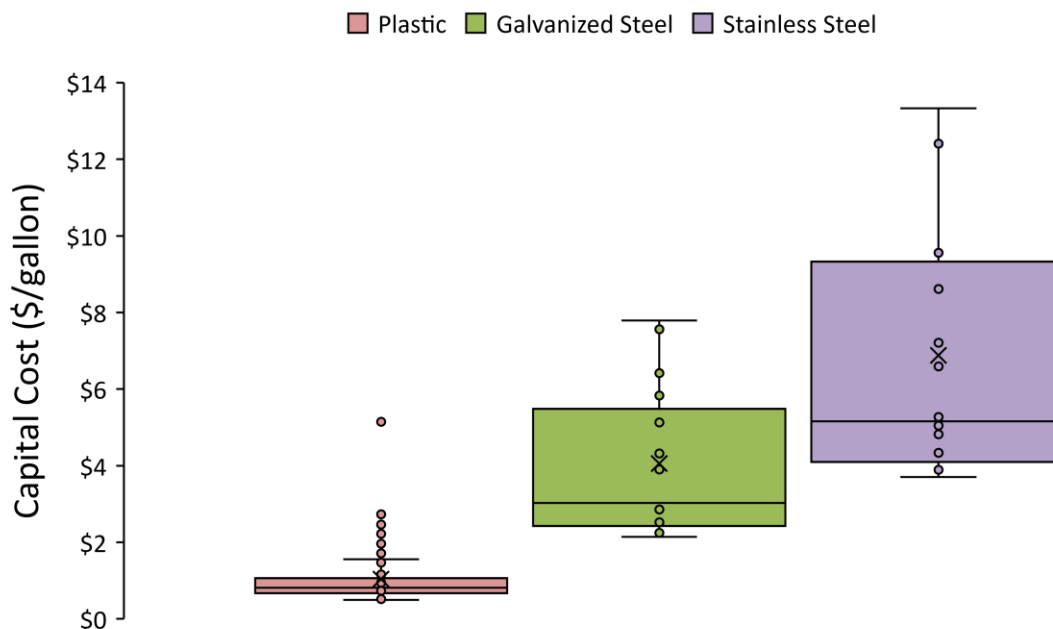


Figure 3-16. Summary of cistern capital costs (plastic, galvanized steel, and stainless-steel).



Table 3-2 Capital costs for bioretention, porous pavement, and cisterns

| BMP Type                                        | Capital Cost            | Source                                                                                                                                                               |
|-------------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bioretention <sup>1</sup>                       | \$207 / ft <sup>2</sup> | Ballona Creek WMG 2015; City of LA 2013, 2017a, 2017b, 2017c, 2017d, 2017f, 2017g, 2019; City of San Marcos 2006; LACDPW 2019b; Malibu Creek WMG 2015; ULAR WMG 2015 |
| Biofiltration <sup>2</sup><br>(w/ Underdrain)   | \$225 / ft <sup>2</sup> |                                                                                                                                                                      |
| Porous Pavement <sup>3</sup>                    | \$253 / ft <sup>2</sup> | City of LA 2017h, 2017i, 2019; City of LA ND; County of San Diego 2017; Newell <i>et al.</i> 2013                                                                    |
| Porous Pavement<br>(w/ Underdrain) <sup>4</sup> | \$265 / ft <sup>2</sup> |                                                                                                                                                                      |
| Cistern, Plastic                                | \$0.8 / gal             | E-commerce Retailer 2019                                                                                                                                             |
| Cistern, Galvanized Steel                       | \$3.0 / gal             |                                                                                                                                                                      |
| Cistern, Stainless Steel                        | \$5.2 / gal             |                                                                                                                                                                      |

1. Using WMMS 2.0 default parameters: 1-ft ponding, 3-ft media depth w/ 0.45 porosity.

2. Using WMMS 2.0 default parameters: 1-ft ponding, 3-ft media depth w/ 0.45 porosity, 1-ft underdrain w/ 0.4 void space.

3. Using WMMS 2.0 default parameters: 3-ft reservoir depth w/ 0.4 porosity.

4. Using WMMS 2.0 default parameters: 3-ft reservoir depth w/ 0.4 porosity, and 1-ft underdrain w/ 0.4 void space.

Cost assumptions for infiltration wells and distributed sewer diversions were derived from information compiled directly from vendors or implementors, as described below:

- ▼ **Infiltration wells:** Capital costs were provided by LACDPW (2019a) based on experience gained through extensive implementation of infiltration wells throughout the County.
- ▼ **Diversion to sewers:** Capital costs for sewer diversions are a composite of two representative projects: (1) a subsurface storage in a small-scale corrugated high-density plastic unit, and (2) an apparatus installed to divert fairly limited flow rates (e.g., 5 cfs) to a sanitary sewer line.

The implementation depth of both of the above BMPs can vary considerably by site (the primary driver for overall cost), therefore costs are characterized on a dollar per cubic foot basis for scalability to various conditions and depths (Table 3-3).

Table 3-3 Capital costs for infiltration wells and diversions to sewer

| BMP Type                        | Capital Cost                      | Source                                   |
|---------------------------------|-----------------------------------|------------------------------------------|
| Infiltration Well <sup>1</sup>  | \$131 / ft <sup>3</sup>           | LACDPW 2019a                             |
| Diversion to Sewer <sup>2</sup> | \$10 / ft <sup>3</sup> + \$67,000 | LACDPW 2019c, Vendor Communication 2019b |

1. Using WMMS 2.0 default parameters: 50 ft reservoir depth, 4-ft inner diameter, 1-ft gravel sleeve w/ 0.4 void space.

2. Per cubic foot cost is for storage volume. Lump sum cost is for sewer connection, associated piping, excavation, disposal, etc. Does not include necessary pump cost (see Section 3.3.2.3) or sanitary sewer discharge fees.

### 3.3.1.2 Operations and Maintenance Costs

To operate at a consistent level of performance over their lifetime, BMPs are required to be appropriately maintained (Taguchi *et al.* 2020). Like capital costs, annual O&M costs vary across a wide range. To account for this range, O&M costs are summarized as a percent of total project capital cost that scales with project size. Descriptions of considerations for O&M costs for each BMP are detailed below. Table 3-4 lists the default O&M costs assigned to each BMP, including data sources used to base these assumptions.

- ▼ **Bioretention.** Journal articles and programmatic documentation were used for annual O&M costs as a percent of the total project cost per square foot. Findings from these sources were averaged and applied to the median costs from Table 3-2. O&M services generally include labor costs related to maintaining vegetation, irrigation, pruning, etc.
- ▼ **Porous pavement.** Two sources were used to calculate annual O&M as a percent total of the project capital cost. O&M services generally include operating a “regenerative air” street sweeper to remove particulate material responsible for fouling the percolative property of the porous material.
- ▼ **Cisterns.** Cisterns generally require limited O&M; therefore, annual O&M costs are not provided as part of this documentation.
- ▼ **Infiltration well.** Information from a regional vendor knowledgeable of the fabrication and installation of infiltration wells was used as the source for annual O&M. Their annual maintenance cost was applied on a per cubic foot basis. O&M services generally include the use of a vacuum truck to remove accumulated debris and sediment in the unit’s pretreatment unit.
- ▼ **Diversion to Sewer.** Similar to infiltration wells, distributed units that divert stormwater to the sewer are installed with a debris trap that must be periodically cleared via a vacuum truck. It was assumed the frequency of this activity for a diversion to sewer would be similar to an infiltration well; therefore, the same O&M costs are applied on a per cubic foot basis.

Table 3-4 Annual O&amp;M costs for distributed BMPs

| BMP Type                      | Annual O&M Cost          | Source                                                          |
|-------------------------------|--------------------------|-----------------------------------------------------------------|
| Bioinfiltration               | \$14.4 / ft <sup>2</sup> | Houle <i>et al.</i> 2013; USEPA 1999; Weiss <i>et al.</i> 2005; |
| Biofiltration                 | \$15.6 / ft <sup>2</sup> |                                                                 |
| Porous Pavement               | \$5.1 / ft <sup>2</sup>  | City of LA 2019; Houle <i>et al.</i> 2013                       |
| Porous Pavement w/ Underdrain | \$5.1 / ft <sup>2</sup>  |                                                                 |
| Infiltration Well             | \$6.6 / ft <sup>3</sup>  | Vendor Communication 2019d                                      |
| Diversion to Sewer            | \$6.6 / ft <sup>3</sup>  |                                                                 |

### 3.3.1.3 Life-Cycle Cost Functions

Using the unit capital costs and O&M costs from above, full life-cycle cost functions were developed for each distributed BMP type (Table 3-5).

Table 3-5 Distributed BMP life-cycle cost functions

| BMP Type           | Life-cycle Cost Function                                                                    |
|--------------------|---------------------------------------------------------------------------------------------|
| Bioretention       | \$207 (A) + \$14.4 (A) (Yr)<br>with Optional underdrain: + \$18 (A) + \$1.2 (A) (Yr)        |
| Porous Pavement    | \$253 (A) + \$5.1 (A) (Yr)<br>Optional underdrain: \$12 (A)                                 |
| Cistern (plastic)  | \$0.8 (Vg)<br>Galvanized steel option: + \$1.2 (Vg)<br>Stainless steel option: + \$4.4 (Vg) |
| Infiltration Well  | \$131 (Vf) + \$6.6 (Vf) (Yr)                                                                |
| Diversion to Sewer | \$10 (Vf) + \$67,000 + \$6.6 (Vf) (Yr)                                                      |

**A** = project surface area in square feet; **Yr** = number of years for annual O&M costs; **Vg** = volume capacity in gallons; **Vf** = volume capacity in cubic feet.

### 3.3.2 Regional BMP Costs

Regional projects can contain an array of components (e.g., diversion structure, pretreatment, storage) with various configurations (on-site irrigation, sewer diversion). For this reason, they are costed by component, configuration, and a per unit volume capacity. The following subsections detail costs for the (1) pretreatment and diversion structure, (2) storage and treatment, and (3) additional configuration options that can be implemented into regional BMP design.

#### 3.3.2.1 Pretreatment and Diversion Structure Costs

Pretreatment and diversion structures are typical front-end components of regional BMPs, as described in Section 3.2. The range of capital costs and associated reference projects are summarized in Figure 3-17. The figure includes box and whisker plots summarizing the range, mean, and median of capital costs for pretreatment and diversion structures included in each project's design.

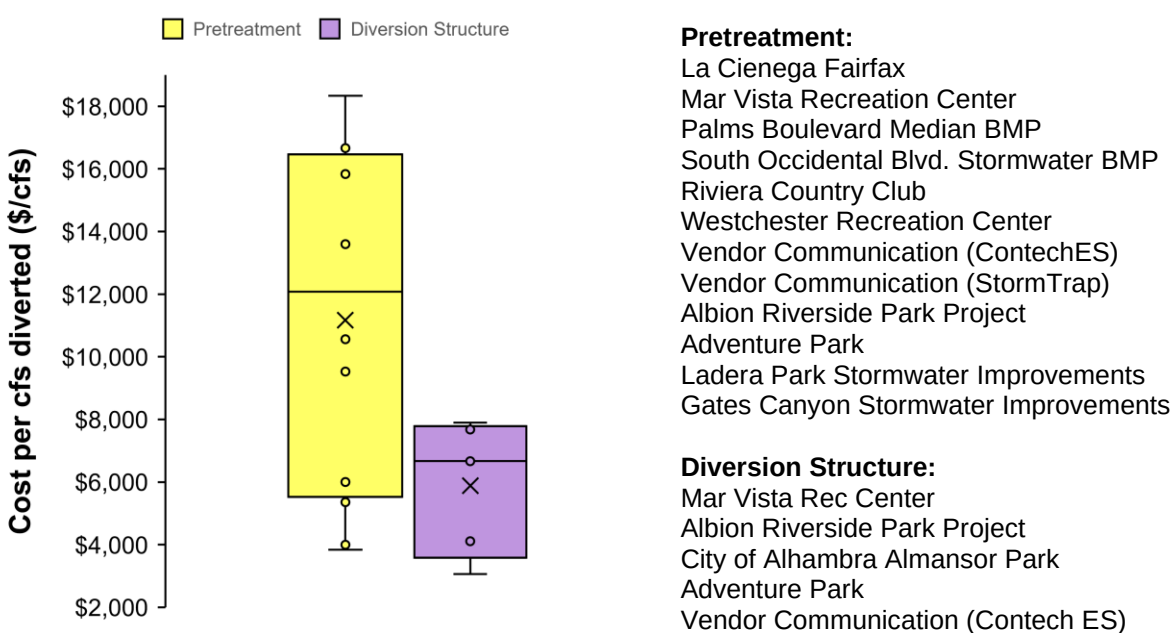


Figure 3-17. Summary of capital costs for pretreatment and diversion structure units.

Median values for capital costs of pretreatment and diversion structure units (Figure 3-17), O&M costs, and associated data sources are summarized in Table 3-6 below. O&M activities for pretreatment and diversion structure units are lumped together (represented in grey in Table 3-6) and generally consist of a vacuum truck used to periodically remove accumulated debris and sediment in the unit's debris trap.

Table 3-6 Regional BMP diversion structure and pretreatment unit capital and O&M costs

| BMP Component       | Capital and O&M (grey) Costs              | Source                                                                                                                                   |
|---------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Diversion Structure | \$7,000 / cfs                             | City of Alhambra 2018; City of LA 2006b, 2013, 2017k; LACDPW 2018a, 2018b; 2019c                                                         |
| Pretreatment Unit   | \$12,000 / cfs (\$320 / cfs) <sup>1</sup> | City of LA 2006a, 2006b, 2006c, 2013, 2015, 2016b, 2017j; LACDPW 2018a, 2018b, 2018c, 2018d, 2019c; Vendor Communication 2019b and 2019c |

1. O&M costs for diversion structures are included in the pretreatment O&M cost.



### 3.3.2.2 Storage and Treatment Costs

Also pivotal to regional BMP design is storage and treatment type. Costs for storage are driven primarily by whether the BMP is sited at the surface or the subsurface. Descriptions of each storage option and the associated costs of implementation and O&M are detailed below and summarized in Table 3-7.

- ▼ **Subsurface infiltration.** As illustrated in the fact sheets above, large-scale subsurface units generally consist of a precast modular concrete system. Information from vendors was used to inform their tabulated cost. For annual O&M costs, cost information from local projects was referenced.
- ▼ **Subsurface filtration.** Subsurface filtration units are typically precast modular concrete systems and utilize the same storage costs as subsurface infiltration units. This project type has an additional cost for the filter units. A major regional vendor was referenced for the cost of treatment on a per-flow basis.
- ▼ **Diversion to sewer.** Diversions to sewer units are typically precast modular concrete systems and utilize the same storage costs as subsurface infiltration/filtration units. This project type has an additional cost for the construction and material needed to discharge stormwater to the sewer network (excavation, shoring, backfilling, and soil removal for pipe installation, material, manhole, and flap gate, among others), presented as a lump sum from a local project. Note, additional fees are generally required to divert stormwater to the sanitary sewer and are not included in this analysis. Pump(s) are also required for this configuration (cost are detailed in the section below).
- ▼ **Surface retention/detention.** Municipal price lists were used for costs associated with implementing surface detention (excavation, soil disposal, riser structure). Journal articles and programmatic documentation comprised the sources accounting for annual O&M costs as a percent of the total project cost per square foot.
- ▼ **Constructed wetland.** Costs from the Proposition O Project, South Los Angeles Wetland, were used for wetland costs. Journal articles and programmatic documentation were used for annual O&M costs as a percent of the total project cost per square foot.

**Table 3-7 Regional BMP storage and treatment unit capital and O&M costs by type**

| BMP Type   |                                  | Storage Unit and O&M (grey) Costs                     | Treatment Capital and O&M (grey) Costs                                    | Source                                                                                                            |
|------------|----------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Subsurface | Infiltration <sup>1, 2, 3</sup>  | \$12.6 / ft <sup>3</sup><br>(\$6,000 / year)          | N/A                                                                       | City of LA 2006a, 2006b; LACDPW 2019c; Vendor Communication 2019a, 2019b, 2019c                                   |
|            | Filtration <sup>1, 2, 3</sup>    |                                                       | \$70,000 / cfs <sup>5</sup><br>Cartridge Filter<br>(\$1,000 / cfs / year) |                                                                                                                   |
|            | Sewer Diversion <sup>1, 2</sup>  |                                                       | \$67,000 <sup>5, 6</sup>                                                  | LACDPW 2019c                                                                                                      |
| Surface    | Detention/Retention <sup>4</sup> | \$15 / ft <sup>2</sup><br>(\$0.6 / ft <sup>2</sup> )  | N/A                                                                       | City of San Marcos 2006; County of San Diego 2017; Houle <i>et al.</i> 2013; USEPA 1999; Weiss <i>et al.</i> 2005 |
|            | Constructed Wetland              | \$112 / ft <sup>2</sup><br>(\$9.1 / ft <sup>2</sup> ) |                                                                           | City of LA, 2006d, 2019; Houle <i>et al.</i> 2013; USEPA 1999; Weiss <i>et al.</i> 2005                           |

1. Precast reinforced concrete vault.

2. Can be combined with optional on-site irrigation (see section below).

3. Can be combined with optional sewer discharge.

4. Using WMMS 2.0 default parameters: 5-ft ponding.

5. Does not include pumping costs (see section below).

6. Does not include sanitary sewer discharge fees.

### 3.3.2.3 Additional Configurations

Regional BMPs can require additional componentry for their operation (e.g., pumps for diversion to sanitary sewers). Descriptions and assumed costs for each additional component are characterized below and summarized with associated data sources in Table 3-8.

- ▼ **On-site irrigation.** Subsurface regional BMPs are often sited beneath parks or other open spaces. In these instances, stormwater stored in the storage facility can be treated on-site and used to irrigate surface vegetation. The cost of installing an irrigation sprinkler system (mainline, laterals, valves, heads, controllers, backflow, etc.) are presented as \$2.20 per square foot of irrigated area plus \$1.1 million for the on-site water treatment system. Treatment systems can include filters and ultraviolet radiation, among other treatment options. These costs were estimated by averaging costs from local projects with on-site treatment. Pump(s) are required for this configuration but are not included in cost assumptions.
- ▼ **Low-flow pump.** Due to their implementation below ground, subsurface regional BMP projects often utilize pumps to lift stormwater. Uses include pumping treated stormwater to the nearby receiving waterbody, the sanitary sewer system, or the surface for irrigation. Pump costs were developed from a broad range of sources including municipal price lists, journal articles, and local projects.

**Table 3-8 Regional BMP component capital and O&M costs**

| Configuration     | Capital and O&M (grey) Costs                   | Source                                                                         |
|-------------------|------------------------------------------------|--------------------------------------------------------------------------------|
| Onsite Irrigation | \$1,100,000 + \$2.20 / sf<br>(\$10,000 / year) | City of LA 2006a, 2006b; LACDPW 2019c                                          |
| Low-flow pump     | \$177,000 / cfs<br>(\$3,000 / cfs / year)      | City LA 2013, 2017j; City of Alexandria 2015; LACDPW 2019c; MAG 2001; PWD 2009 |

### 3.3.2.4 BMP Life-Cycle Cost Functions

The unit costs for pretreatment/diversion, storage/treatment, and additional components, as described above, were used to develop the full life-cycle cost functions for regional BMPs (Table 3-9). These regional BMP cost functions and those listed for distributed projects in Table 3-5 provide the foundation for assessing the cost-effectiveness of relative BMP implementation strategies in the WMMS 2.0 Optimization Framework (Section 4.3).

**Table 3-9 Regional BMP life-cycle costs functions**

| BMP Type            | Life-cycle Cost Function                                                                                                                                                                                                                                                         |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dry Detention       | \$19,000 (In) + \$15 (A) + \$0.6 (A)(Yr) + \$320 (In)(Yr)                                                                                                                                                                                                                        |
| Wetland             | \$19,000 (In) + \$112 (A) + \$9.1 (A)(Yr) + \$320 (In)(Yr)                                                                                                                                                                                                                       |
| Regional Subsurface | $\$19,000 (In) + \$12.6 (Vf) + \$6,000 (Yr) + \$320 (In)(Yr)$<br>Filter <sup>1</sup> : + \$247,000 (Eff) + \$4,000 (Eff) (Yr)<br>Irrigation: + \$1,100,000 + \$177,000(Pi) + \$3,000(Pi)(Yr) + \$2.2(A) + \$10,000(Yr)<br>Sewer: + \$67,000 + \$177,000 (Ps) + \$3,000 (Ps) (Yr) |

1. Assumes pump on BMP effluent.

**A** = project surface area in square feet; **Yr** = number of years for annual O&M costs; **Vg** = capacity volume in gallons; **Vf** = capacity volume in cubic feet; **In** = inflow rate to the BMP in cfs; **Eff** = effluent rate from the BMP in cfs; **Pi** = Pump flow rate to on-site irrigation in cfs; **Ps** = Pump flow rate to sewer in cfs.

## 4 WMMS 2.0 WEB-BASED MODELING PLATFORM

Los Angeles County is a diverse area containing the most densely populated urban area in the country (US Census 2018), expansive chaparral landscapes in its headwaters (Homer *et al.* 2004), and a complex hydraulic system (e.g., engineered channels and diversions) that serve to convey and redirect stormwater (LACFCD 2011b). The WMMS 2.0 Phase I Report describes in detail how these aspects were incorporated into the LSPC model for hydrology and water quality calibration and validation. This section describes how the calibrated runoff and constituent loading from LSPC can serve as a boundary condition to SUSTAIN, using the suite of utilities and viewers integrated into the WMMS 2.0 web-based modeling platform. Identifying cost-optimal suites of BMP locations, footprints, and capacities in a highly developed area like Los Angeles (Yang *et al.* 2018) is an essential aspect of evaluating and developing meaningful and impactful BMP implementation strategies. Following development of the original WMMS applications (LACDPW 2010a, 2010b, 2011), common feedback on the optimization elements was lack of transparency on modeling assumptions, challenges in the use of the optimization framework to inform watershed management decisions, and the difficulty in managing data and file structure across the modeling workflow. The utilities in WMMS 2.0 address these concerns and provide a more approachable and capable modeling and optimization process to further empower strategic watershed management in the Los Angeles region.

The three primary components of the WMMS 2.0 web platform are the Model and Timeseries Repository, Network and Timeseries Viewer, and Optimization Framework, described below.

- ▼ **Model and Timeseries Repository.** The calibrated LSPC model (described in the WMMS 2.0 Phase I Report) and the SUSTAIN model (described in this report) are made available for download in the Repository. The Repository also includes executables for each model as well as the meteorological input files for LSPC.
- ▼ **Network and LPSC Timeseries Viewer.** The LSPC model domain includes an extensive network of waterbodies, channels, flood control facilities, and subwatersheds. To illustrate how this hydraulic modeling network is integrated into the LSPC model and various model inputs and outputs, an interactive map and data viewer has been developed.
- ▼ **Optimization Framework.** The Optimization Framework consists of a series of utilities that simulate and evaluate existing and proposed BMPs to meet user-identified targets (e.g., 22% reduction in zinc load) across a watershed management area (e.g., Upper Los Angeles River). BMPs are selected via a two-tiered optimization approach leveraging SUSTAIN's NSGA-II algorithms to identify a set of cost-effective options. To facilitate this Framework, a consistent stepwise structure of utilities and an Optimization Viewer have been developed to guide users through the process.
  - **WRAMPS Utility.** Existing BMPs can have a significant impact on downstream hydrology and water quality. If applicable, users can initiate the Optimization Framework by downloading data from the Watershed Reporting, Adaptive Management and Planning System (WRAMPS) database. WRAMPS is a key repository for data regarding existing BMPs in the County.
  - **BMP Opportunities Utility.** Output from the WRAMPS Utility is a spreadsheet template where existing BMPs from WRAMPS are organized in a SUSTAIN model-ready format. This spreadsheet also provides a template for proposed BMPs to be characterized (e.g., dimensions, drainage area, costs) for simulation. The completed spreadsheet can be uploaded to the BMP Opportunities Utility whose output is a set of SUSTAIN input files containing all the necessary model details for existing and proposed BMPs needed to initiate the first tier of the optimization.



- **Inclusive Best Solutions Utility.** Output from the first tier of the optimization can be uploaded to the Inclusive Best Solutions Utility. This utility sorts and organizes the BMP cost-effectiveness curves from each “jurished” (combination of model subwatersheds and jurisdictional boundaries represented in the WMMS 2.0 model domain, see Section 4.3 for more detail), ensuring that incremental solutions are inclusive of previous ones. Output from this Utility is used to initiate the second tier of optimization.
- **MJR BMP Utility.** If multi-jurished regional (MJR) BMPs are present, output from the second tier of the optimization can be uploaded to the MJR BMP Utility. MJR BMPs are regional projects whose drainage areas include more than one jurished. To incorporate these facilities into the modeling framework, the utility organizes results from the two-tiered optimization for jurisheds upstream of MJR BMPs as a hydrologically connected network to be evaluated during optimization.
- **Two-tiered Optimization Utility.** Output from the MJR BMP Utility (if MJR BMPs are considered) or the second tier of optimization (if MJR BMPs are not considered) can be uploaded to the Two-tiered Optimization Utility. This Utility creates a file matrix that summarizes BMP capacity, reduction, and cost from all optimization runs in a format that can be uploaded to the Optimization Viewer for interactive analysis of model results.
- **Optimization Viewer.** Output from the Two-tiered Utility can be uploaded directly to the Optimization Viewer. The Viewer enables the simulated BMP configurations, corresponding cost, and modeled effectiveness to be plotted to assess performance. The user can navigate through solutions along the curve to see how BMP capacities, responses, and costs vary. Each solution on the curve is a unique implementation plan that can be selected and exported using the viewer.

Figure 4-1 illustrates the Optimization Framework as the progression from problem formulation, through the utilities, and ultimately to the Optimization Viewer. As described above, each aspect of the Optimization Framework supports SUSTAIN input file generation, two-tiered model simulation, BMP performance evaluation, and implementation planning.

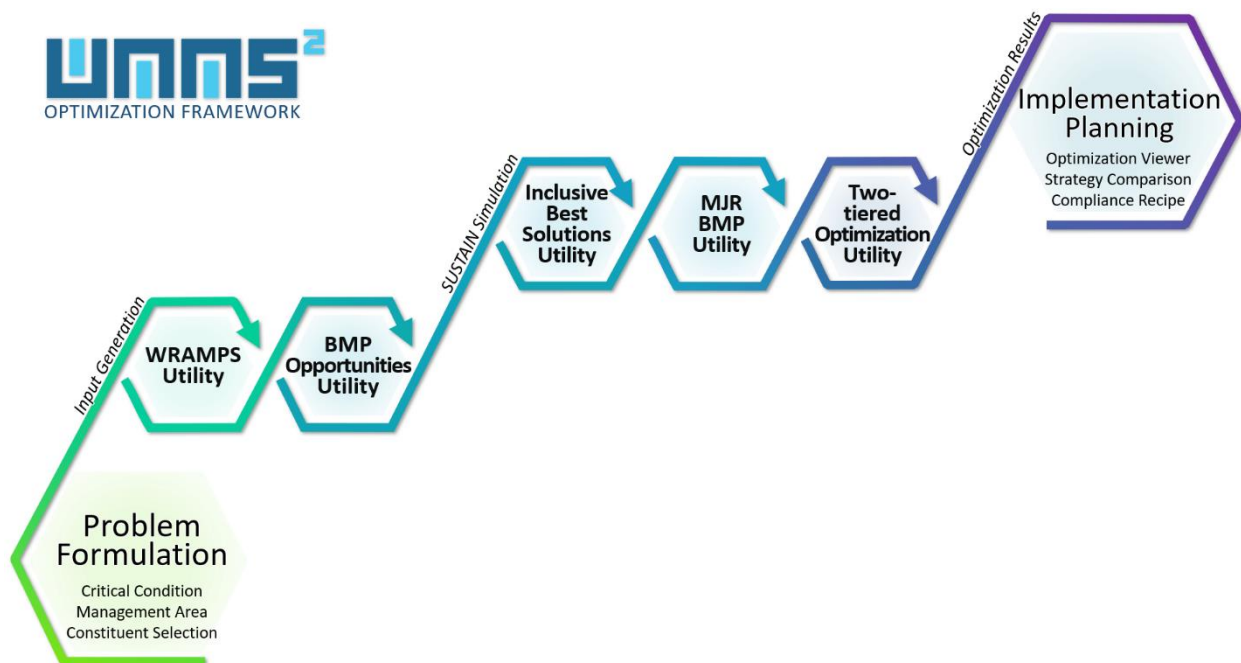


Figure 4-1. Conceptual process map of the WMMS 2.0 Optimization Framework utilities and viewer.

## 4.1 Model and Timeseries Repository

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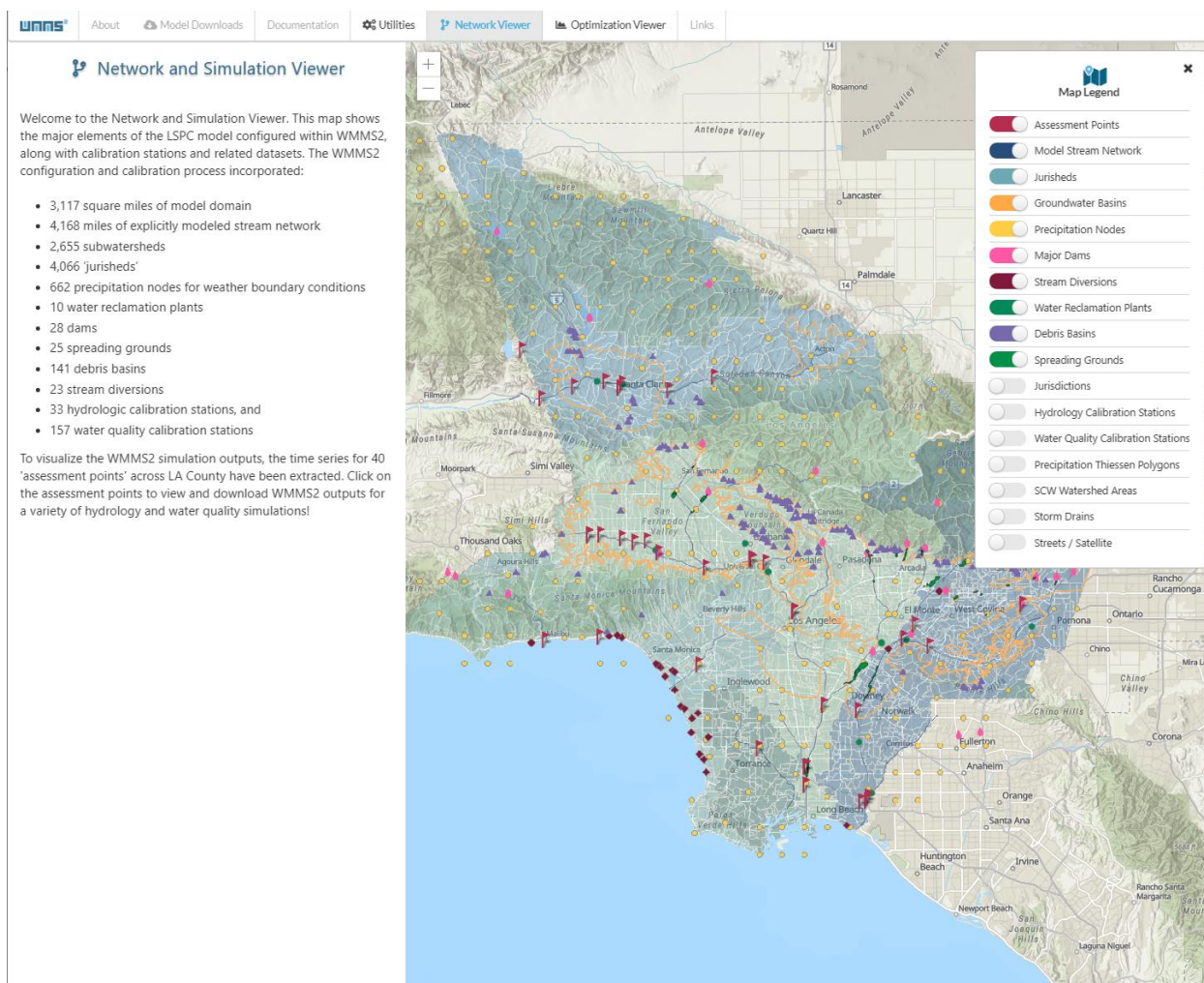
The LSPC and SUSTAIN model configurations, their executables, and LSPC meteorological input files (precipitation and evapotranspiration) are available for download in the Model and Timeseries Repository. Collectively, these items represent a user's launching pad for initiating model simulation and serve as a consistent, defensible starting point for future RAAs. The meteorological files represent a continuous timeseries that drive runoff and pollutant loading processes in LSPC. Meteorological files for the 10-year calibration and validation period and other critical conditions for evaluating BMP performance (e.g., 90<sup>th</sup> percentile critical water year, 85<sup>th</sup> percentile storm depth) are also made available:

- ▼ **10-year continuous timeseries.** Input files representing precipitation and evapotranspiration from water year 2009 to 2018 comprise the LSPC calibration and validation timeseries. These timeseries were developed from the LACDPW's Automatic Local Evaluation in Real Time (ALERT) rainfall gage network and incorporated gridded meteorological products from the Parameter-elevation Regressions on Independent Slopes Model (PRISM) and the North American Land Data Assimilation System-2 (NLDAS) to interpolate gaps in spatial coverage. The WMMS 2.0 Phase I Report should be referenced for additional details on the development of these timeseries.
- ▼ **Critical conditions.** Four water quality conditions, as determined from applicable total maximum daily loads (TMDLs), E/WMPs, and other documentation, are available for download. They include the 85<sup>th</sup> percentile 24-hour storm, critical bacteria storm, and identification of average and 90<sup>th</sup> percentile water years across major watersheds in the Los Angeles region. To describe how each of these conditions were characterized, a "Simulation Document" has been developed and is made available on the Repository.

## 4.2 Network and Timeseries Viewer

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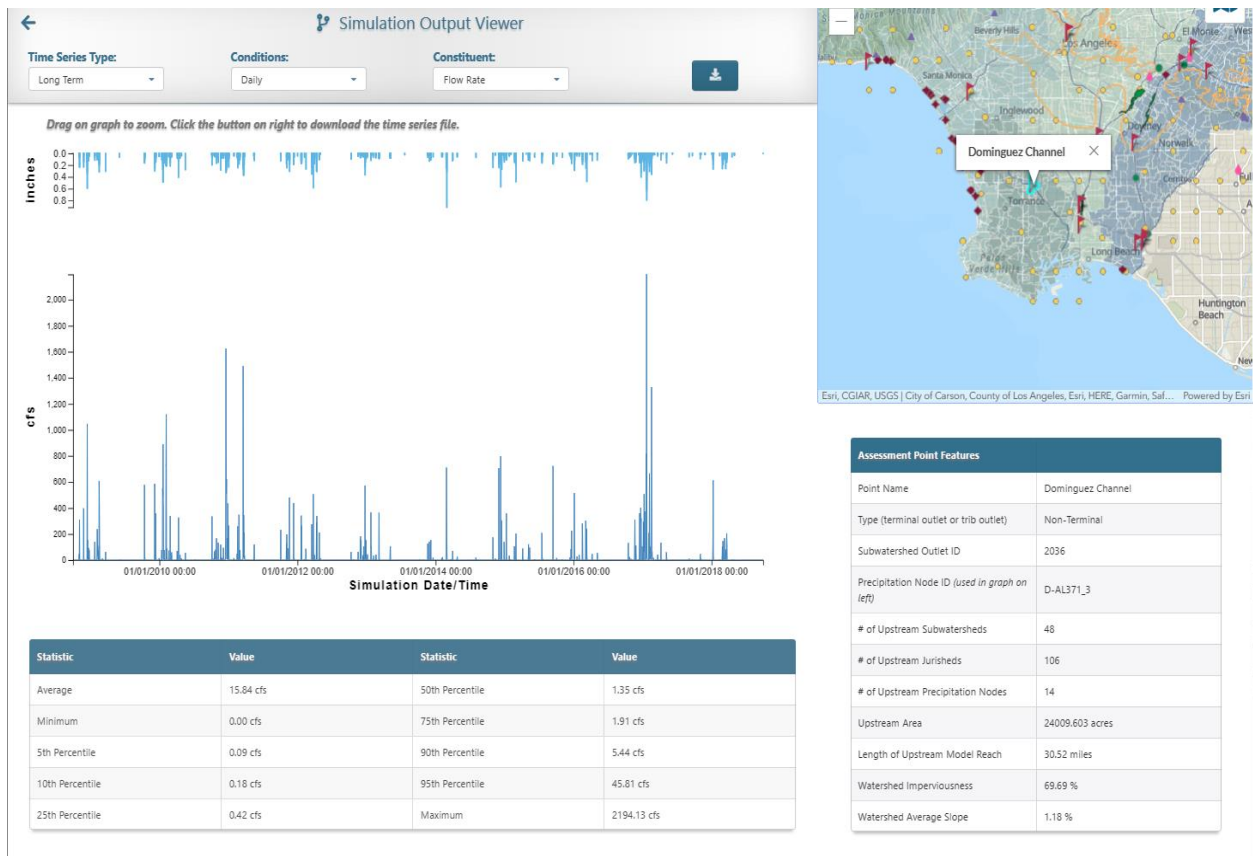
The Network and Timeseries Viewer was developed to illustrate the location and details of the datasets utilized in the LSPC model. Datasets available in the Viewer's interactive map include meteorological inputs (ALERT stations and PRISM and NLDAS nodes), hydraulic considerations (e.g., waterbodies, channels, flood control facilities), and other relevant layers (e.g., city boundaries) across the model domain. Figure 4-2 shows the Viewer's current landing page.



**Figure 4-2. WMMS 2.0 Network and LSPC Timeseries Viewer with selected model items selected.**

Within the Network Viewer map, approximately 40 assessment points are available for viewing and exporting detailed model output. Output includes model attributes (e.g., upstream drainage area), summary statistics (e.g., median flow), as well as the LSPC precipitation, runoff, and instream pollutant concentration timeseries. Figure 4-3 illustrates the Dominguez Channel assessment point as an example with the 10-year modeled runoff and precipitation timeseries plotted.



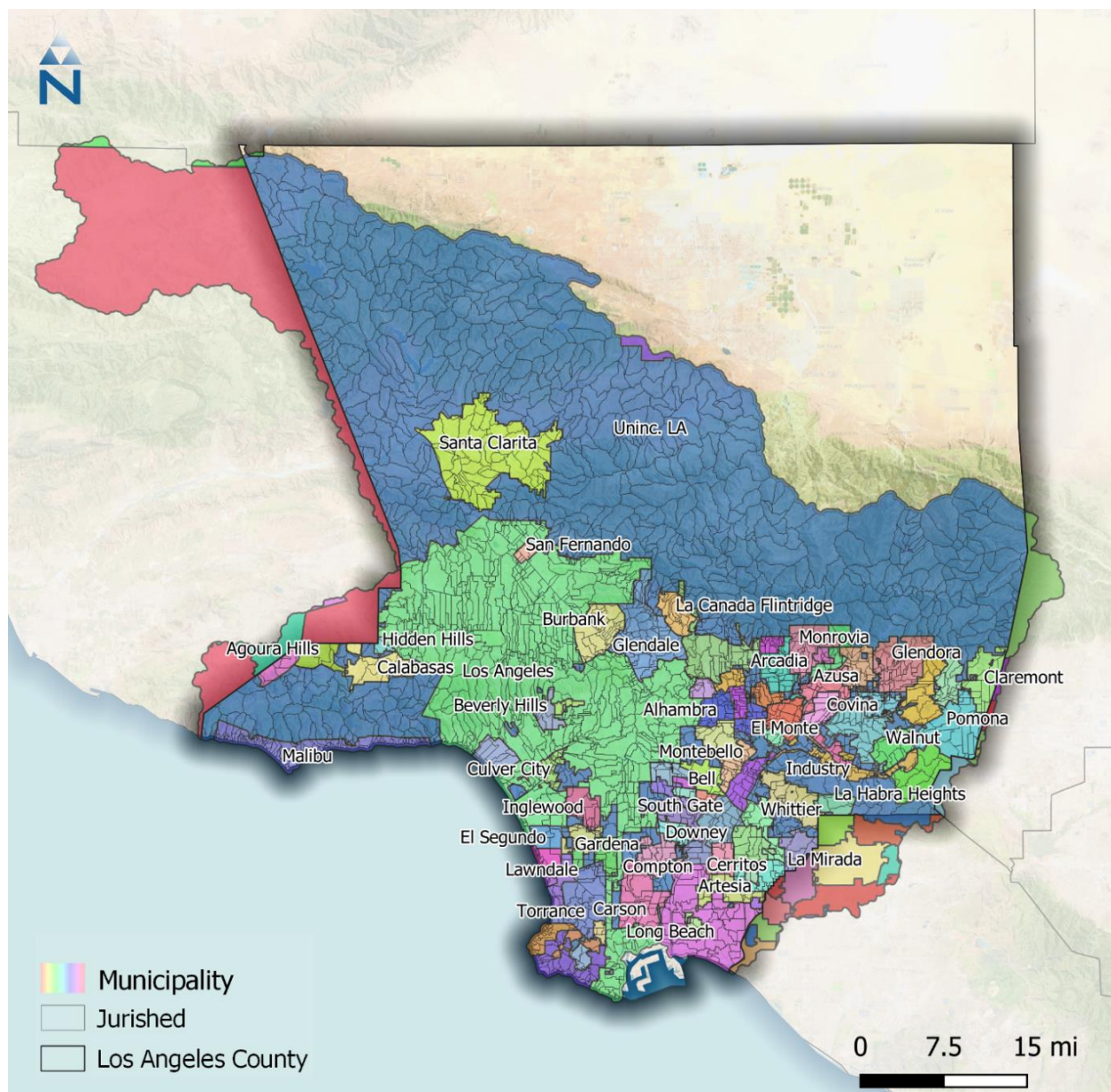


**Figure 4-3 Flow and precipitation for the Dominguez Channel point in the Network and LSPC Timeseries Viewer.**

From the Timeseries Viewer, users can select the timeseries to evaluate and download for each of the 40 assessment points. The Network and Timeseries Viewer provides transparency for model input, characterization, and the flexibility to view, evaluate, and download region-specific runoff and water quality data. Once the downloaded LSPC model has been further calibrated as needed, the Optimization Framework can be initiated to evaluate and organize cost-effective BMPs.

### 4.3 Optimization Framework

Model simulations in the Optimization Framework are organized at the subwatershed or jurished scale. The jurished construct allows modeled volume and loads to be summarized by jurisdictional boundaries while maintaining hydrologic connectivity. Jurisheds across the model domain were developed by intersecting the 2,655 subwatersheds from the LSPC model with the most recent municipal jurisdictional boundaries (LAC GIS Portal 2019). Because jurisdictional and hydrologic boundaries do not coincide perfectly, the intersection of the two layers occasionally results in instances of especially small jurisheds, some as small as 500 square feet. To control for small jurisheds, a lower boundary set by the smallest LSPC subwatershed area (1.74 acres) was enforced (i.e., jurisheds smaller than 1.74 acres after the intersection were merged into the most proximal largest neighbor). This process resulted in 4,139 distinct jurisheds (Figure 4-4).

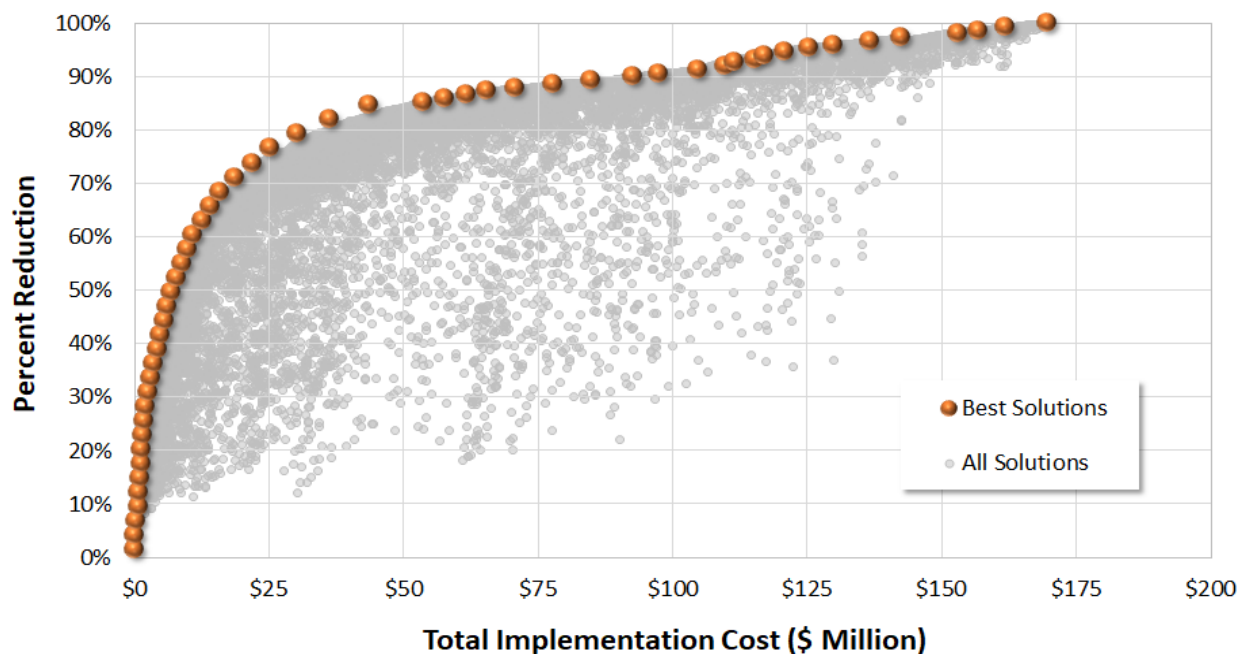


**Figure 4-4. Jurished boundaries with major city labels across the model area.**

Model organization at the jurished scale means that existing and proposed BMP opportunities and their respective parameters (e.g., costs, drainage area) are summarized in a single SUSTAIN input file per jurished. This structure enables users to focus their modeling efforts on the runoff and opportunities within their jurisdictional boundaries and then summarize output for their jurisdiction.

To assess existing and proposed BMP types and volumes in this model organization, a two-tiered optimization was developed. In all considered jurisheds, a cloud of thousands (sometimes tens of thousands) of unique BMP combinations described by cost and performance are generated; this is the first tier of the optimization. Each BMP combination is a distinct blend of BMP types and configurations that represents a potential BMP implementation plan for that jurished. Example output from a jurished is shown in Figure 4-5. Each gray dot in the plot (“all solutions”) is an evaluated BMP combination in the cloud. The larger orange “best solutions” points are the most cost-effective

solutions within the jurisdiction—this curve represents the highest achievable performance at every cost interval.



**Figure 4-5. Example solutions for a single jurisdiction and the advantage of cost-benefit optimization.**

Each jurisdiction's set of best solutions represents the most cost-effective options within its boundary. However, when compared with results from another jurisdiction, it is sometimes more cost-effective on a regional scale to manage more aggressively in one jurisdiction to meet downstream objectives. The second tier of optimization is used to evaluate sets of best solutions from multiple jurisdictions for a management objective at a downstream point. This process balances the varying cost and benefits of all considered best solutions across the larger management area to identify a cost-optimal BMP strategy for each level of pollutant removal to meet downstream objectives (as determined by the user). Most E/WMPs used this two-tiered approach to develop regional cost-effectiveness curves that synthesized results from multiple jurisdictions. On the left-hand side of Figure 4-6, a conceptual graphic of "N" number of jurisdictions and their corresponding best solutions curves are illustrated. Each of these Tier 1 curves corresponds to a jurisdiction outlet; but the Tier 2 optimization curve comprises all the upstream Tier 1 curves. Figure 4-6 illustrates Tier 2 as two layers of optimization; however, conceptually, whenever MJR BMPs (routed/nested features) are added to the network, a third layer of optimization is created.



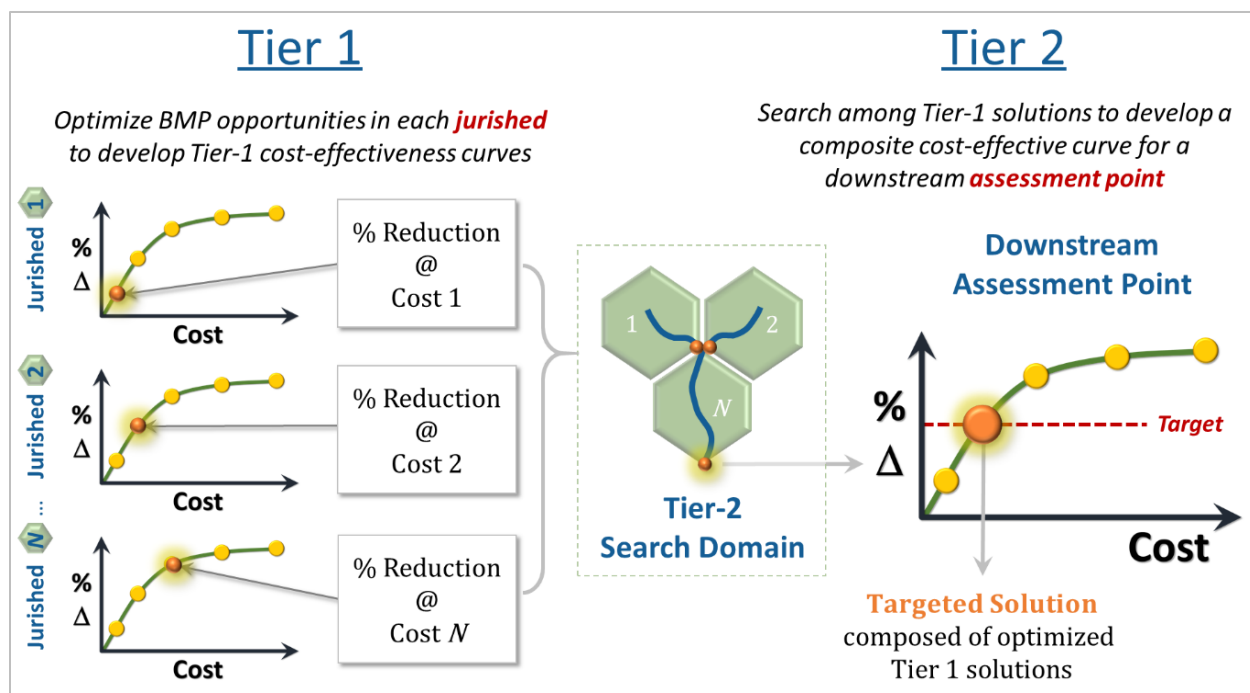


Figure 4-6. Conceptual representation of the two-tiered optimization approach.

### 4.3.1 WRAMPS Utility

WRAMPS is a web-based application developed and maintained by LACFCD for MS4 reporting that includes tracking of structural BMPs and estimates of their effectiveness. WRAMPS uses a combination of GIS data, watershed modeling, and web technologies to provide users with visualizations of water capture, reports to illustrate MS4 compliance, and an inventory of each municipality's existing and planned projects. Municipalities that use WRAMPS for annual reporting purposes can access that system to enter, modify, and download existing projects in the database for use in the WRAMPS Utility. Other users can enter existing projects manually and initiate the Optimization Framework with the BMP Opportunities Utility described in the next section.

Once within WRAMPS, users have the option to download the complete set of BMPs currently cataloged within the database via the bulk download option. This download summarizes all cataloged projects and their corresponding attributes (e.g., location, project volume) in a detailed Excel spreadsheet format within their managed area .

The WRAMPS Utility accepts this bulk download file and translates its contents into model parameters that can be simulated in SUSTAIN. Specifically, output from the WRAMPS Utility is a spreadsheet template containing all uploaded WRAMPS BMPs characterized for model simulation. Users can also enter proposed BMPs they have identified and characterized separately in a screening analysis. This process enables a user to develop input files that consider both existing BMPs from WRAMPS *and* proposed BMP in the same model environment for optimization. (further described in Section 4.3.2 BMP Opportunities Utility).

The specific attributes used to characterize existing BMPs in the WRAMPS Utility from WRAMPS are:

- ▼ **Location.** BMP locations are described in WRAMPS by their latitude and longitude, enabling the corresponding jurished to be determined. Jurished determination allows each existing BMP and

their modeling variables (e.g., dimensions, upstream drainage area) to be organized into the appropriate input file.

- ▼ **BMP type (LID, green street, or regional) and storage capacity.** WRAMPS does not require a user to input the explicit physical dimensions (footprint and depth) of a BMP, however both are necessary input variables for SUSTAIN. Because WRAMPS does require each BMP's "project storage capacity" and "BMP type" to be entered, approximations of BMP footprint and depth are made by the WRAMPS Utility. First, the utility assigns each BMP type from WRAMPS an equivalent type in the WMMS 2.0 Menu of BMPs. Then, the default depths associated with each BMP type, as defined in the menu (Section 3.1), are used to divide the project storage capacity from WRAMPS to calculate a footprint. This dimensional translation between BMPs in WRAMPS to those in the WMMS 2.0 Menu of BMPs is summarized for each BMP in Table 4-1.

**Table 4-1. Translation from BMP output in WRAMPS to BMP representation in the WRAMPS Utility**

| WRAMPS Output              |                                                 | WRAMPS Utility Characterization  |                     |                              |                                                                                                                                                                                                                                               |
|----------------------------|-------------------------------------------------|----------------------------------|---------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BMP Type and Configuration |                                                 | WMMS BMP Type                    | Utility ID          | Effective Depth <sup>1</sup> | BMP Footprint                                                                                                                                                                                                                                 |
| LID                        | Biofiltration                                   | Biofiltration                    | exLID1              | 2.75                         | BMP footprints are back calculated by dividing the storage capacity provided in the WRAMPS database by the assumed effective depth for each BMP (column to the left).<br><br>$\text{BMP Footprint} = \frac{\text{Volume}}{\text{Eff. depth}}$ |
|                            | Onsite Infiltration                             | Bioretention                     | exLID1              | 2.35                         |                                                                                                                                                                                                                                               |
|                            | Rainfall Harvesting                             | Bioretention                     | exLID3              | 2.35                         |                                                                                                                                                                                                                                               |
| Green Street               | Green Street (filtration) w/ porous pavement    | Biofiltration w/ porous pavement | exLID4 w/ exLID8    | 2.75 w/ 1.6 <sup>2, 3</sup>  |                                                                                                                                                                                                                                               |
|                            | Green Street (infiltration) w/ porous pavement  | Bioretention w/ porous pavement  | exLID2 w/ exLID6    | 2.35 w/ 1.6 <sup>2, 3</sup>  |                                                                                                                                                                                                                                               |
|                            | Green Street (filtration) w/o porous pavement   | Biofiltration                    | exLID3              | 2.75                         |                                                                                                                                                                                                                                               |
|                            | Green Street (infiltration) w/o porous pavement | Bioretention                     | exLID1              | 2.35                         |                                                                                                                                                                                                                                               |
|                            | Porous pavement                                 | Porous Pavement                  | exLID5              | 1.6 <sup>2</sup>             |                                                                                                                                                                                                                                               |
| Regional                   | Infiltration Facility                           | Infiltration Facility            | Varies <sup>4</sup> | 10                           |                                                                                                                                                                                                                                               |
|                            | Treatment Facility                              | Filtration Facility              |                     | 10                           |                                                                                                                                                                                                                                               |
|                            | Diversion to Sewer                              | Diversion to Sewer               |                     | 10                           |                                                                                                                                                                                                                                               |

1. Effective depth. Takes into account ponding height, media depth, void space, etc.

2. Effective depth for the porous pavement aspect of the green street.

3. To model bioretention with porous pavement, routing in the Utility output is predefined to ensure these two BMPs are linked (i.e., nested) in the model appropriately.

4. Regional facilities are given ID's of for the order in which they appear in the WRAMPS database (i.e., the first listed regional BMP is "exREG1". See the "Lumping" subsection below.

- ▼ **Drainage area.** A SUSTAIN simulation requires an explicit drainage area representation of the upstream HRUs. HRUs are discrete elements that make up the hydrologic landscape (e.g., commercial impervious) and are used to characterize water quality and quantity of runoff in the

model. A full description of model HRUs is provided in the WMMS 2.0 Phase I Report. The following summarizes how HRUs are determined for categories of BMPs included in WRAMPS and those regional BMPs represented in the MJR BMP Utility (Section 4.3.1).

- *Green Streets and LID*: WRAMPS does not contain a detailed HRU drainage area breakdown, only their total area and imperviousness. To generate HRU breakdowns, the WRAMPS Utility proportionally distributes HRUs for the jurisdiction in which the BMP is located, to the drainage area of the BMP. In other words, the overall percent breakdown of HRUs at the jurisdiction-scale are presumed as the relative HRU breakdown at the BMP drainage-scale—the sum of the area will also match the drainage area acreage provided in WRAMPS. Note that if WMMS users have the specific breakdown of HRUs within a BMP drainage area, they can overwrite the default WRAMPS Utility output with their data.
- *Regional BMPs*: WRAMPS has recently been updated to store drainage area delineation and HRU representations for regional BMPs. To be represented in the Utility process, all regional BMPs must have an uploaded drainage area (e.g., .shp format). If one is not uploaded, a user will have to delineate the HRU proportions and enter them manually.
- *MJR BMPs*: Drainage area delineations and HRU breakdowns are not provided for MJR BMPs and must be entered manually. See the *BMP Opportunity Utility User Document* for additional details.
- ▼ **BMP Lumping**. LID and green street BMP types from WRAMPS are collectively represented in WMMS as LID (“exLID1” through “exLID6” as shown in Table 4-1). When more than one of any LID type is present in a jurisdiction, they are represented as a *lumped* BMP. For example, if there are 500 existing biofiltration units in jurisdiction 100449, their footprint and drainage area are lumped into “exLID1”. Existing regional BMPs are distinct in that they can only be modeled individually, and their footprint and drainage areas cannot be lumped. This approach is necessary because regional projects often have unique configurations (e.g., diversion structure, sewer diversion, filtration unit) that cannot be sufficiently modeled using a lumped approach. Therefore, when values for maximum footprint for regional BMP projects are entered into the BMP size tab, they should be representative of a single project for each type (as many as 50 regionals and 50 MJR BMPs).

### 4.3.2 BMP Opportunities Utility

The BMP Opportunities Utility translates existing BMPs from the WRAMPS database *and* proposed BMPs from a screening analysis (performed separately as part of a planning process) into model-ready input files. Existing and proposed BMPs are characterized in the BMP opportunities spreadsheet template (spreadsheet template) described in the previous section, which is an output from the WRAMPS Utility. If the WRAMPS Utility was not used, a blank spreadsheet template is available for download on the BMP Opportunities Utility webpage. Five tabs are contained within the spreadsheet template to characterize BMPs by the evaluation factors (e.g., selected pollutants), design, footprint, and drainage area. The following subsections describe the importance of each of these tabs in the model process.

#### Simulation Configuration

The Simulation Configuration tab includes options for each input file that exports from the BMP Opportunities Utility (Figure 4-7) and are described below.

- ▼ **Simulation begin/end date**. Month, day, and year when the simulation begins/ends. These dates will vary based on the selected timeseries (e.g., 10-year calibration period, 90<sup>th</sup> percentile water year, 85<sup>th</sup> percentile storm event).
- ▼ **Evaluation factor**. SUSTAIN can be set to evaluate one of several different constituents in the optimization. Depending on management area drivers, users can select an evaluation factor that



the optimization algorithm will use to assess BMP performance. All constituents characterized in the WMMS 2.0 Phase I Report are available for assessment in a dropdown menu. They include:

- Average annual flow volume (cubic feet/year).
- Annual average load for total suspended solids, total nitrogen, total phosphorous, total cadmium, total copper, total lead, or total zinc (pounds per year).
- ▼ **Maximum Iterations.** This defines the number of iterations before an optimization is terminated and should be large enough to converge on a solution. The more decision variables (i.e., proposed BMPs) within a watershed, the greater this value should be. Minimum suggested value is 10,000 for 1 to 5 BMPs. If all possible BMP types are simulated, as many as 30,000 iterations or more may be required to converge on a solution.
- ▼ **Management levels.** This value is the number of solutions (at equal intervals) sampled from the Tier 1 optimization curve to be simulated in the Tier 2 optimization process. In other words, the number of points used to make up the best solutions curve selected from the cloud of tens of thousands of solutions from Tier 1 to be used as input for Tier 2 optimization.

| Simulation Start Date | Simulation End Date | Evaluation Factor | Maximum Iterations | Management Levels |
|-----------------------|---------------------|-------------------|--------------------|-------------------|
| 10/1/2017             | 9/30/2018           | Flow Volume       | 15,000             | 100               |

Figure 4-7. Simulation Configuration tab in the BMP opportunities spreadsheet template.

## Design Details

The Design Details tab contains all BMPs types from the menu and their corresponding parameters (e.g., cost, porosity) as established in Section 3. The preassigned default parameter values can also be modified by a user to match specifications of a set of projects or areas or allow for user preferences. BMP type is the primary driver for organization in this tab. Four broad categories have been designated and are described below.

- ▼ **Existing LID and Regional (REG).** BMPs as determined and characterized from the WRAMPS database. As many as 10 existing LIDs (“exLID1” through “exLID10”) and 50 existing regionals (“exREG1” through “exREG50”).
- ▼ **Proposed LID and REG.** New BMP opportunities from a separate screening process. As many as 20 proposed LIDs (“LID1” through “LID20”) and 10 proposed regionals (“REG1” through “REG10”) can be characterized by configuration (e.g., diversion rate, infiltration, downstream routing). BMP lumping for proposed BMPs is consistent with how existing BMPs are characterized in the WRAMPS Utility. When more than one of any of the LID types is present in a watershed, their footprint and drainage area are lumped. Proposed regional BMPs are distinct in that they are modeled individually, and their footprint and drainage areas are not lumped. The drainage areas for each REG BMP should be contained within its corresponding watershed. If any REG BMP contains portions of more than one watershed, it should be represented as a MJB BMP.
- ▼ **Proposed MJB BMPs.** A regional project whose drainage area is contained within more than one watershed is considered a MJB BMP. As many as 50 MJB BMP types can be simulated in a single watershed-scale optimization run.
- ▼ **Proposed TBD BMP.** In instances where the maximum footprints from the existing and proposed BMPs are unable to meet the specified water quality targets, a “TBD BMP” option is provided. This BMP type is not a physically located opportunity. Rather, it is a way to assess how much

additional BMP capture volume would be required to meet the specified water quality target. In other words, if during the two-tiered optimization process, no possible combinations of BMPs can meet the user-specified load reduction, the TBD BMP can be leveraged into the process as a proxy for use in setting goals for additional BMPs (yet to be identified) to meet targets. With this type of consideration included in an optimization, SUSTAIN is able to assess the balance between the cost and benefit of implementing this conceptual TBD BMP opportunity against existing and proposed BMPs. As such, its parameterization (i.e., determination of size, drainage area, routing, cost) is extremely important for representing the trade-offs between feasible opportunities and yet-to-be-defined opportunities. Not all optimizations require a TBD BMP to be defined; therefore, it is included as an option a user can exercise when appropriate.

Figure 4-8 illustrates a possible set of BMP types and corresponding parameters in the BMP Size tab. Note that listed BMP types can be duplicated as many times as desired by a user, up to the maximum for each type. For example, in Figure 4-8 five different regional projects are listed for modeling (infiltration, filtration, diversion to sewer, detention, and wetland). A user can change this to set all five regional BMPs to infiltration facilities if desired, or any combination of BMP quantity and type (up to a maximum of 10 for regional BMPs).

| BMP ID and Description |             |                    |                                           | BMP Parameters and Cost   |                |             |            |          |         |                      |                     |                   |                         |
|------------------------|-------------|--------------------|-------------------------------------------|---------------------------|----------------|-------------|------------|----------|---------|----------------------|---------------------|-------------------|-------------------------|
| ID                     | WMMS BMP ID | Modeled BMP Type   | BMP Description                           | Optimization Viewer Group | Diversion Rate | Weir Height | Soil Depth | POROSITY | FINFILT | Vegetative Parameter | Holtan Growth Index | Underdrain Switch | Underdrain Infiltration |
| ID                     | BMPID       | BMP TYPE           | DESCRIPTION                               | LABEL                     | DIVRATE        | WEIRH       | SDEPTH     | POROSITY | FINFILT | AVEG                 | GI                  | UNDSWITCH         | UNDFINFILT              |
| Proposed LID           | 1 LID1      | BIORETENTION       | Bioinfiltration                           | LID, Bioretention         | -1             | 1           | 3          | 0.45     | 5       | 0.8                  | 0                   | 1 NoUnderdrain    | 1.5                     |
|                        | 2 LID2      | RAINBARREL         | Cistern                                   | LID, Other                | -1             | 8           | 0          | 0.45     | 5       | 0                    | 0                   | 0 NoUnderdrain    | 1.5                     |
|                        | 3 LID3      | POROUSPAVEMENT     | Porous Pavement                           | LID, Other                | -1             | 0.001       | 3          | 0.40     | 5       | 0                    | 0                   | 0 NoUnderdrain    | 1.5                     |
|                        | 4 LID4      | INFILTRATIONTRENCH | Infiltration Well                         | LID, Other                | -1             | 62.05       | 0          | 0.40     | 5       | 0                    | 0                   | 0 NoUnderdrain    | 1.5                     |
|                        | 5 LID5      | DRYPOND            | Diversion to Sewer                        | LID, Other                | -1             | 4           | 0          | 0.45     | 5       | 0                    | 0                   | 0 NoUnderdrain    | 1.5                     |
|                        | 6 LID6      | DRYPOND            | Treatment Unit                            | LID, Other                | -1             | 4           | 0          | 0.45     | 5       | 0                    | 0                   | 0 NoUnderdrain    | 1.5                     |
| Proposed REG           | 100 REG1    | INFILTRATIONTRENCH | Infiltration Facility                     | Regional Infiltration     | -1             | 10          | 0.5        | 0.40     | 5       | 0                    | 0                   | 0 NoUnderdrain    | 1.5                     |
|                        | 101 REG2    | DRYPOND            | Treatment Facility - Filtration           | Regional Treatment        | -1             | 10          | 0          | 0.45     | 5       | 0                    | 0                   | 0 NoUnderdrain    | 1.5                     |
|                        | 102 REG3    | DRYPOND            | Treatment Facility - Sewer Diversion      | Regional Treatment        | -1             | 10          | 0          | 0.45     | 5       | 0                    | 0                   | 0 NoUnderdrain    | 1.5                     |
|                        | 103 REG4    | WETPOND            | Treatment Facility - Wetland              | Regional Treatment        | -1             | 5           | 2          | 0.45     | 5       | 0.8                  | 1                   | 1 NoUnderdrain    | 1.5                     |
|                        | 104 REG5    | DRYPOND            | Treatment Facility - Detention            | Regional Treatment        | -1             | 5           | 2          | 0.45     | 5       | 0.3                  | 1                   | 1 NoUnderdrain    | 1.5                     |
| Proposed MJR           | 999 TBD BMP | DRYPOND            | To Be Determined                          | Regional TBD              | -1             | 10          | 0.5        | 0.40     | 5       | 0                    | 0                   | 0 NoUnderdrain    | 1.5                     |
|                        | 200 MJRBMP1 | INFILTRATIONTRENCH | Infiltration Facility                     | MJR BMP                   | -1             | 10          | 0.5        | 0.40     | 5       | 0                    | 0                   | 0 NoUnderdrain    | 1.5                     |
|                        | 201 MJRBMP2 | INFILTRATIONTRENCH | Infiltration Facility                     | MJR BMP                   | -1             | 10          | 0.5        | 0.40     | 5       | 0                    | 0                   | 0 NoUnderdrain    | 1.5                     |
|                        | 202 MJRBMP3 | INFILTRATIONTRENCH | Infiltration Facility                     | MJR BMP                   | -1             | 10          | 0.5        | 0.40     | 5       | 0                    | 0                   | 0 NoUnderdrain    | 1.5                     |
|                        | 203 MJRBMP4 | INFILTRATIONTRENCH | Infiltration Facility                     | MJR BMP                   | -1             | 10          | 0.5        | 0.40     | 5       | 0                    | 0                   | 0 NoUnderdrain    | 1.5                     |
| Existing LID           | 1000 exLID1 | BIORETENTION       | Bioinfiltration                           | Existing LID              | -1             | 1           | 3          | 0.45     | 5       | 0.8                  | 1                   | 1 NoUnderdrain    | 1.5                     |
|                        | 1003 exLID2 | BIORETENTION       | Biofiltration (Routed to Porous Pavement) | Existing LID              | -1             | 1           | 3          | 0.45     | 5       | 0.8                  | 1                   | 1 NoUnderdrain    | 1.5                     |
|                        | 1004 exLID3 | POROUSPAVEMENT     | Porous Pavement                           | Existing LID              | -1             | 0.001       | 3          | 0.4      | 5       | 0                    | 0                   | 0 NoUnderdrain    | 1.5                     |
|                        | 1008 exLID4 | INFILTRATIONTRENCH | Infiltration Well                         | Existing LID              | -1             | 62.05       | 0          | 0.4      | 5       | 0                    | 0                   | 0 NoUnderdrain    | 1.5                     |
|                        | 1009 exLID5 | DRYPOND            | Diversion to Sewer                        | Existing LID              | -1             | 4           | 0          | 0.45     | 5       | 0                    | 0                   | 0 NoUnderdrain    | 1.5                     |
| Existing REG           | 1010 exREG1 | BIORETENTION       | -                                         | Existing Regional         | -1             | 1           | 3          | 0.45     | 5       | 0.8                  | 1                   | 1 NoUnderdrain    | 1.5                     |
|                        | 1011 exREG2 | BIORETENTION       | -                                         | Existing Regional         | -1             | 1           | 3          | 0.45     | 5       | 0.8                  | 1                   | 1 NoUnderdrain    | 1.5                     |
|                        | 1012 exREG3 | BIORETENTION       | -                                         | Existing Regional         | -1             | 1           | 3          | 0.45     | 5       | 0.8                  | 1                   | 1 NoUnderdrain    | 1.5                     |
|                        | 1013 exREG4 | BIORETENTION       | -                                         | Existing Regional         | -1             | 1           | 3          | 0.45     | 5       | 0.8                  | 1                   | 1 NoUnderdrain    | 1.5                     |
|                        | 1014 exREG5 | BIORETENTION       | -                                         | Existing Regional         | -1             | 1           | 3          | 0.45     | 5       | 0.8                  | 1                   | 1 NoUnderdrain    | 1.5                     |

Figure 4-8. Design Details tab in the BMP opportunities spreadsheet template.

## BMP Size

For each proposed BMP entered into the Design Details tab, a maximum BMP footprint is entered in the BMP Size tab. Values should originate from a separate user-completed screening analysis, which identifies potential BMP opportunities (existing LID and REG BMPs will be already populated if the WRAMPS Utility was used). As shown in Figure 4-9, this tab is organized with jurisdictions as rows and the BMP types as columns.

|    | A         | B    | C      | D    | E     | F    | G     | H    | I    | J    | K     | L     | M     | N     | O     | P     | Q     | R     | S     | T     | U     | V    | W    | X    | Y    | Z    | AA   |
|----|-----------|------|--------|------|-------|------|-------|------|------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|------|------|------|------|
| 1  | JURISHEID | LID1 | LID2   | LID3 | LID4  | LID5 | LID6  | LID7 | LID8 | LID9 | LID10 | LID11 | LID12 | LID13 | LID14 | LID15 | LID16 | LID17 | LID18 | LID19 | LID20 | REG1 | REG2 | REG3 | REG4 | REG5 | REG6 |
| 2  | 100149    | 450  | -      | -    | 2,200 | -    | 1,100 | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -    | -    | -    | -    | -    |      |
| 3  | 100183    | -    | 120    | -    | -     | 450  | 360   | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -    | -    | -    | -    | -    |      |
| 4  | 100249    | -    | 15,000 | -    | 750   | -    | -     | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -    | -    | -    | -    | -    |      |
| 5  |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 6  |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 7  |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 8  |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 9  |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 10 |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 11 |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 12 |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 13 |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 14 |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 15 |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 16 |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 17 |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 18 |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 19 |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 20 |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 21 |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 22 |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 23 |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 24 |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 25 |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |
| 26 |           |      |        |      |       |      |       |      |      |      |       |       |       |       |       |       |       |       |       |       |       |      |      |      |      |      |      |

Figure 4-9. BMP Size tab in the BMP opportunities spreadsheet template.

## Drainage Area

For each proposed BMP entered in the Design Details tab, the breakdown of HRUs upstream of each BMP type is entered in the Drainage Area tab (Figure 4-10). For each HRU, the total area of all columns must equal the area of the HRU in each jurished. To maintain area continuity, HRU areas that do not drain to a BMP should be tabulated in the “Untreated” column. Reference the *BMP Opportunities Utility User Document* for a detailed description of how HRU areas are determined.

|    | A         | B     | C                              | D         | E    | F    | G    | H    | I    | J    | K    | L    | M    | N     | O     | P     | Q     | R     | S     | T     | U     | V     | W     | X     | Y    | Z    | AA   |
|----|-----------|-------|--------------------------------|-----------|------|------|------|------|------|------|------|------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|------|
| 1  | JURISHEID | HRU   | HRU-DESC                       | UNTREATED | LID1 | LID2 | LID3 | LID4 | LID5 | LID6 | LID7 | LID8 | LID9 | LID10 | LID11 | LID12 | LID13 | LID14 | LID15 | LID16 | LID17 | LID18 | LID19 | LID20 | REG1 | REG2 | REG3 |
| 2  | 700549    | 1000  | Road_Freeway-All-All-All       | -         | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 3  | 700549    | 2000  | Road_Primary-All-All-All       | 34.0      | 14.6 | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | 2.1   | -     | -     | -     | -     | -     | -     | -     | 4.0  | -    |      |
| 4  | 700549    | 3000  | Road_Minor-All-All-All         | 61.4      | 27.5 | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | 4.2   | -     | -     | -     | -     | -     | -     | -     | 9.0  | -    |      |
| 5  | 700549    | 4000  | Dev_ResHigh-All-All-All        | 0.1       | 0.0  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 6  | 700549    | 5000  | Dev_ResLow-All-All-All         | 81.7      | 41.0 | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | 2.8   | -     | -     | -     | -     | -     | -     | -     | 11.1 | -    |      |
| 7  | 700549    | 6000  | Dev_Com-All-All-All            | 15.8      | 0.2  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | 3.0  | -    |      |
| 8  | 700549    | 7000  | Dev_Ind-All-All-All            | -         | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 9  | 700549    | 8000  | Dev_Inst-All-All-All           | 23.4      | 5.8  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | 0.0   | -     | -     | -     | -     | -     | -     | -     | 2.0  | -    |      |
| 10 | 700549    | 9000  | Dev_Roof-All-All-All           | 137.5     | 67.8 | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | 5.0   | -     | -     | -     | -     | -     | -     | -     | 20.7 | -    |      |
| 11 | 700549    | 10000 | Dev_Overspray-All-All-All      | 6.4       | 2.5  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | 0.2   | -     | -     | -     | -     | -     | -     | -     | 0.8  | -    |      |
| 12 | 700549    | 11111 | Dev_Irrigated-A-Low-Confined   | 1.0       | 0.0  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | 0.4   | -     | -     | -     | -     | -     | -     | -     | 0.0  | -    |      |
| 13 | 700549    | 11112 | Dev_Irrigated-A-Low-Unconfined | 12.0      | 7.6  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | 0.2   | -     | -     | -     | -     | -     | -     | -     | 3.3  | -    |      |
| 14 | 700549    | 11121 | Dev_Irrigated-A-Med-Confined   | 0.1       | 0.0  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | 0.0   | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 15 | 700549    | 11122 | Dev_Irrigated-A-Med-Unconfined | 0.1       | 0.2  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | 0.0  | -    |      |
| 16 | 700549    | 11211 | Dev_Irrigated-B-Low-Confined   | 0.0       | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 17 | 700549    | 11212 | Dev_Irrigated-B-Low-Unconfined | 4.4       | 3.7  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 18 | 700549    | 11221 | Dev_Irrigated-B-Med-Confined   | 0.3       | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 19 | 700549    | 11222 | Dev_Irrigated-B-Med-Unconfined | 0.1       | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 20 | 700549    | 11311 | Dev_Irrigated-C-Low-Confined   | 1.2       | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | 0.0   | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 21 | 700549    | 11312 | Dev_Irrigated-C-Low-Unconfined | 1.1       | 0.0  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 22 | 700549    | 11321 | Dev_Irrigated-C-Med-Confined   | 4.3       | 0.0  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | 0.2   | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 23 | 700549    | 11322 | Dev_Irrigated-C-Med-Unconfined | 1.0       | 0.2  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 24 | 700549    | 11411 | Dev_Irrigated-D-Low-Confined   | 0.1       | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | 0.0   | -     | -     | -     | -     | -     | -     | -     | 0.0  | -    |      |
| 25 | 700549    | 11412 | Dev_Irrigated-D-Low-Unconfined | 0.5       | 0.3  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | 0.1   | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 26 | 700549    | 11421 | Dev_Irrigated-D-Med-Confined   | 0.9       | 0.0  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 27 | 700549    | 11422 | Dev_Irrigated-D-Med-Unconfined | 0.5       | 0.0  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | 0.1   | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 28 | 700549    | 12111 | Dev_Pervious-A-Low-Confined    | 2.5       | 0.1  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | 1.1   | -     | -     | -     | -     | -     | -     | -     | 0.0  | -    |      |
| 29 | 700549    | 12112 | Dev_Pervious-A-Low-Unconfined  | 30.9      | 19.5 | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | 0.5   | -     | -     | -     | -     | -     | -     | -     | 8.4  | -    |      |
| 30 | 700549    | 12121 | Dev_Pervious-A-Med-Confined    | 0.3       | 0.1  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | 0.0   | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 31 | 700549    | 12122 | Dev_Pervious-A-Med-Unconfined  | 0.2       | 0.4  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | 0.0  | -    |      |
| 32 | 700549    | 12211 | Dev_Pervious-B-Low-Confined    | 0.1       | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 33 | 700549    | 12212 | Dev_Pervious-B-Low-Unconfined  | 11.3      | 9.5  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 34 | 700549    | 12221 | Dev_Pervious-B-Med-Confined    | 0.9       | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 35 | 700549    | 12222 | Dev_Pervious-B-Med-Unconfined  | 0.2       | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 36 | 700549    | 12311 | Dev_Pervious-C-Low-Confined    | 3.2       | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | 0.1   | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 37 | 700549    | 12312 | Dev_Pervious-C-Low-Unconfined  | 2.9       | 0.0  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |
| 38 | 700549    | 12321 | Dev_Pervious-C-Med-Confined    | 11.1      | 0.1  | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | 0.5   | -     | -     | -     | -     | -     | -     | -     | -    | -    |      |

Figure 4-10. Drainage Area tab in the BMP opportunities spreadsheet template.

## MJR BMP Drainage

Areas draining to MJR BMPs are characterized in the MJR BMP Drainage tab. To streamline the process, this tab should be filled with the percent of each jurished overlapped by each MJR BMP's drainage area as shown in Figure 4-11. The entered percentage proportionally samples each jurished from the Drainage Area tab and applies it in the SUSTAIN model to the listed MJR BMP. The partitioning does not specify *which* HRUs are routed. Instead, flow and pollutant load from the jurished is proportionally routed to the corresponding MJR BMP.



|     | A            | B        | C             | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T |
|-----|--------------|----------|---------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1   | JURISDICTION | MJR BMP  | DRAINAGE AREA |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 2   | 106506       | MJR BMP1 | 100%          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 3   | 106406       | MJR BMP1 | 33%           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 15  | 123306       | MJR BMP2 | 80%           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 16  | 124906       | MJR BMP2 | 78%           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 17  | 124806       | MJR BMP2 | 38%           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 203 |              |          |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 204 |              |          |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 205 |              |          |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 206 |              |          |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 207 |              |          |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 208 |              |          |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 209 |              |          |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 210 |              |          |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 211 |              |          |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 212 |              |          |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 213 |              |          |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 214 |              |          |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 215 |              |          |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 216 |              |          |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 217 |              |          |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 218 |              |          |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 219 |              |          |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

Figure 4-11. MJR BMP Drainage tab in the BMP opportunities spreadsheet template.

Once the spreadsheet template has been completed, it can be uploaded to the BMP Opportunities Utility. Output from the BMP Opportunities Utility is a set of SUSTAIN input and batch files in a specific organization of folders to expedite the optimization process. The *BMP Opportunities Utility User Document* should be referenced for specific instructions on how to execute the first tier of the optimization on a user's desktop computer.

### 4.3.3 Inclusive Best Solutions Utility

Once the first tier of the optimization has been successfully executed, outputs are organized and arranged to form an inclusive set of best solutions. As described above, outputs from the first tier of optimization is a cloud of unique BMP combinations for each modeled jurisdiction (all solutions; Figure 4-5). By default, each optimization solution in the cloud is independent of every other solution. During an optimization, BMP performance may favor a certain suite of BMPs at lower reduction intervals and different suite at higher intervals. In practice, it would be cumbersome for long-term implementation planning to implement one suite of BMPs at lower percent reduction, then implement another suite for achieving higher reduction at a later time. Identifying a set of best solutions that is “inclusive” addresses this issue by selecting BMP suites that have the same base BMP configuration. For example, the configuration of BMPs that achieves a 10% reduction will be included in the suite that achieves 50% reduction. The BMP configuration in the 50% reduction suite will contain additional and/or larger BMPs. The Inclusive Best Solutions Utility is a post-process that incrementally combs through the Tier 1 cloud to build a solution set where higher-reduction solutions are inclusive of lower ones. The number of points sampled along the perimeter of the curve is determined by the “management level” variable described in Section 4.3.2. For example, the recommended management level of 100 solutions will produce an inclusive Tier 1 curve of 100 best solutions from the tens of thousands in the cloud.

SUSTAIN output from Tier 1 is uploaded to the Inclusive Best Solutions Utility, which returns an inclusive set of best solutions for each cloud. Once these files have been downloaded the second tier of the optimization can be initiated. The *Optimization Utilities User Document* should be referenced for specific instructions. Once completed, output can be uploaded to the MJR BMP Utility (if MJR BMPs are considered) or the Two-Tiered Optimization Utility (if MJR BMPs are not considered).

### 4.3.1 MJR BMP Utility

When MJR BMPs are simulated within a model area, the MJR BMP Utility will accept output from the Tier 2 optimization. Large regional projects can be especially cost-effective opportunities because of the economies of scale they provide as they treat stormwater from large upstream areas. Because

the Optimization Framework operates at the jurished scale, downstream MJR BMPs are optimized upon completion of the second tier of optimization. This utility organizes output from the second-tier optimization such that they can be evaluated with MJR BMPs that are being optimized in tandem through this process. As needed, reference the *Optimization Utilities User Document* for specific guidance on the execution of this simulation.

### 4.3.2 Two-tiered Optimization Utility

The Two-tiered Optimization Utility organizes and assembles solutions from the two-tier optimization process, with or without the MJR BMP results, to develop a composite regional cost-effectiveness curve for a downstream point. This utility is where problem formulation for watershed-scale optimization occurs. It is the culmination of all the pre-processing utilities and model runs presented thus far. In an optimization, terms like “objective,” “target,” and “scenario” are often used interchangeably. In the context of the Two-tiered Optimization Utility they are part of key terminology associated with different stages of problem formulation, as outlined below:

- ▼ **Management Objective.** The “driver” for BMP optimization. It represents the overarching goal of the optimization. This is defined at both the Tier 1 and Tier 2 levels. Each objective is defined for an evaluation factor (e.g., flow, total zinc, etc.). Example management objectives include:
  - Maximize runoff volume reduction
  - Maximize pollutant load reduction
- ▼ **Assessment Point.** Location where a management objective is evaluated during optimization. Assessment points can be either actual (e.g., hydrologically connected outlet of a routing segment) or virtual (e.g., a sum of jurisheds from one municipality in a broader management area). Either way, results are aggregated at an assessment point and evaluated during optimization.
- ▼ **Decision Variable.** A dimension that changes during optimization (e.g., BMP footprint, diversion rate). The full range of decision variables among all BMP opportunities (as described in Section 4.3.2) represents the search domain for optimization. The number of combinations in the search domain grows exponentially depending on the number of BMPs and step intervals among decision variables.
- ▼ **Management Scenario.** The combination of management objectives, assessment points, and/or decision variables to be evaluated. Varying the management objective (e.g., runoff volume vs. pollutant load) produces different cost-effectiveness curves when all other elements are equal. Likewise, changing decision variables (e.g., using distributed BMPs vs. large regional projects, or a combination of both) also produces different curves when all other elements are equal. Well-formulated management scenarios can provide meaningful results to inform implementation planning efforts.
- ▼ **CRC Matrix.** The cost-effectiveness curve output from the utility is a matrix of BMP capacities (“C”) with associated responses (“R”) and costs (“C”). Each management scenario run produces a unique CRC matrix output. Each solution in the matrix represents a different spatial footprint of BMPs within the jurisheds associated with the assessment point. The CRC matrix is a relatively complex three-dimensional data construct. The BMP capacity, pollutant response, and associated costs all vary spatially between solutions along the curve. The optimization viewer utility (described in Section 4.3.3) provides a way to meaningfully interact with CRC matrices.
- ▼ **Target.** The target is a point along the “R” axis of the CRC matrix that represents the level of performance for the scenario. Depending on available opportunity and how the problem is formulated, the desired target may or may not be achievable (i.e., there may not be a solution in the CRC matrix). Section 4.3.2 described a “TBD” opportunity that may be used to ensure that a solution exists in the output for the desired target.
- ▼ **Implementation Plan.** A “slice” of the CRC matrix associated with a specific target that achieves the intended level of performance for a defined scenario. The plan includes the spatial distribution of capacity for each BMP type by jurished and associated cost for the slice that collectively achieves

the specified target for the scenario. An implementation plan can be generated from the Optimization Viewer, as detailed in the next section.

With the key terminology defined above, two of the utility's primary processes for organizing and assembling solutions into a CRC matrix can be described.

### Jurisdictional and Basin-wide Implementation

It is a common objective to assess the cost and pollutant reduction tradeoffs between jurisdictional and basin-wide BMP implementation strategies across an assessment area. In a jurisdictional implementation approach, each municipality implements a slice from the CRC matrix that represents the most cost-effective BMP implementation option for the jurisdictions within their boundary. However, when results are compared across multiple municipalities in the assessment area, it may be more cost-effective at a basin-wide scale to manage more aggressively in specific municipalities to meet downstream management objectives. For example, soils in one municipal jurisdiction may provide better infiltration than a neighboring jurisdiction, resulting in less BMPs basin-wide if the more effective jurisdictions with higher infiltration are targeted for implementation. For this reason, an alternative basin-wide implementation approach may be useful. For the basin-wide modeling scenario, municipal boundaries are not considered. Instead, all BMP options are assessed within the model domain for maximum reduction of the target constituent at the assessment point. The spatial distribution of implementation cost and capacity is strongly influenced by which implementation strategy is selected. The *Optimization Utilities User Document* describes how the Two-tiered Optimization Utility is configured to assess both jurisdictional and basin-wide strategies.

### MJR BMP Cost and Benefit Sharing

An additional consideration that strongly impacts the spatial distribution of implementation is how MJR BMP costs and benefits are shared among municipal jurisdictions that share the project drainage area. Because the drainage areas of MJR BMPs often overlap multiple municipalities, two options have been developed to “bookend” how their costs and benefits are evaluated. For example, consider the blue MJR BMP in Figure 4-12. This project receives runoff and pollutant loading from three upstream jurisdictions. The cost and benefit of this MJR BMP can either be “split” with the contributing upstream jurisdictions, or it can “stay” and be assigned only to the jurisdiction where the MJR BMP is located. These split/stay options provide a valuable tool to inform how MJR BMPs are incorporated in the Optimization Framework and project costs and benefits are considered in an implementation plan. The red MJR BMP in Figure 4-12 overlaps with only one jurisdiction; therefore, it is not assessed from a split/stay perspective (i.e., cost and benefit are all contained within one jurisdiction). The *Optimization Utilities User Document* describes how the Two-tiered Optimization Utility can be configured to assess MJR BMPs that overlap multiple jurisdictions with the split/stay options.

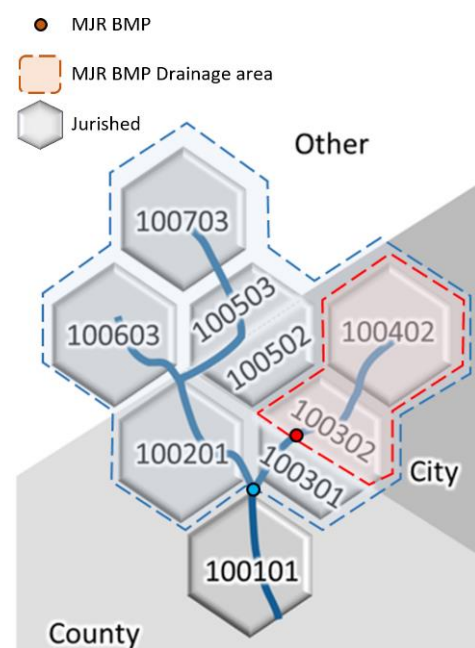


Figure 4-12. Assessment area with two MJR BMPs across three jurisdictions.

### 4.3.3 Optimization Viewer

The Optimization Viewer has been developed to summarize results from the Optimization Framework and allow users to meaningfully interact with those results and compare implementation strategies. Output from the Two-tiered Optimization Utility is uploaded to the Optimization Viewer



and aggregated into a summary plot that illustrates the complete range of constituent reduction by all opportunities in the CRC matrix. Figure 4-13 provides a diagram illustrating the three axes of the summary plot (capacity, reduction, cost) with a solution (i.e., a slice) selected at 33.35 ac-ft capacity, 45% reduction, and \$30 million. Solutions are organized from low to high along the R-axis, aggregated by BMP type as a cumulative volume distribution plot on the primary C-axis, and the associated cost-effectiveness curve as a solid black line on the secondary C-axis. Users can move the vertical slice along the R-axis of the curve to see how cost and capacity change for each point along the plot. Every point represents the optimal solution set that achieves the specified target at the assessment point for the lowest cost. From here, BMP implementation strategies can be selected, compared, and exported into a workbook that can be used to inform watershed and BMP implementation planning efforts.

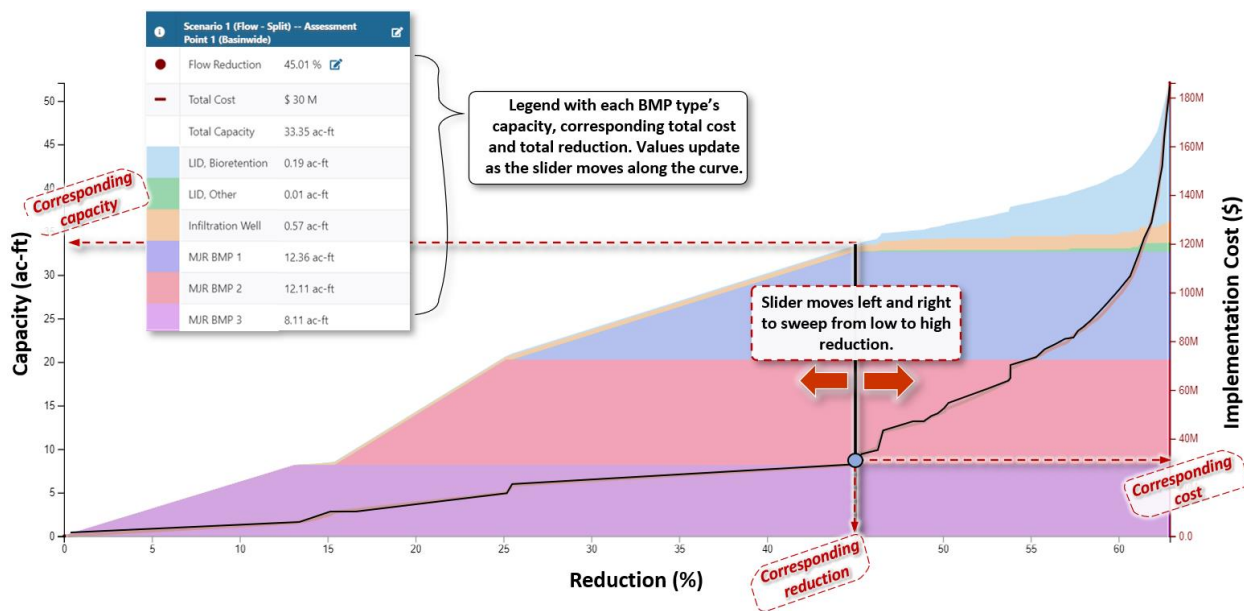


Figure 4-13. Diagram of a summary plot with labels for corresponding capacity, reduction, and costs.

To provide additional functionality and utility, other tools and options have been integrated into the Optimization Viewer to summarize results and inform BMP implementation plan development

- ▼ **Comparison table.** This option enables users to compare summary plots and tables from the CRC outputs of different scenarios. Figure 4-14 provides an example of a single assessment point in the watershed evaluated for three different constituents with various load reductions. This option to simultaneously compare the cost-benefit of alternative implementation strategies would be useful to E/WMP groups interested in assessing the full range of approaches for reducing downstream pollutant load or volume.

| Optimization Metric   | Scenario 2 (SOSED) -- Assessment Point 1 (DS All) | Scenario 1 (Flow) -- Assessment Point 1 (DS All) | Scenario 3 (PLS_TN) -- Assessment Point 1 (DS All) |
|-----------------------|---------------------------------------------------|--------------------------------------------------|----------------------------------------------------|
| Reduction             | Flow: 17.47%                                      | PLS_TP: 18.41%                                   | Flow: 15.77%                                       |
| Total Cost            | \$48 M                                            | \$28 M                                           | \$36 M                                             |
| LID, Bioretention     | 4.95 ac-ft                                        | 3.08 ac-ft                                       | 4.03 ac-ft                                         |
| LID, Other            | 0.35 ac-ft                                        | 0.13 ac-ft                                       | 0.19 ac-ft                                         |
| Infiltration Well     | 0.30 ac-ft                                        | 0.02 ac-ft                                       | 0.03 ac-ft                                         |
| Regional Infiltration | 3.85 ac-ft                                        | 3.76 ac-ft                                       | 3.76 ac-ft                                         |
| Regional Treatment    | 0.00 ac-ft                                        | 0.00 ac-ft                                       | 0.00 ac-ft                                         |
| Regional TBD          | 0.00 ac-ft                                        | 0.00 ac-ft                                       | 0.00 ac-ft                                         |
| Existing LID          | 0.00 ac-ft                                        | 0.00 ac-ft                                       | 0.00 ac-ft                                         |
| Existing Regional     | 0.00 ac-ft                                        | 0.00 ac-ft                                       | 0.00 ac-ft                                         |

Figure 4-14. Comparison table of three implementation strategies selected from the Optimization Viewer.

- ▼ **BMP Implementation Plan Generator.** Once various BMP solutions and configurations have been considered, users can select to develop a BMP implementation plan using the Plan Generator. An implementation plan details the selected BMP type, volume, and cost across each modeled jurisdiction at the desired level of reduction. Output from the Generator is a spreadsheet that summarizes model results by jurisdiction to detail the selected BMP types, volumes, costs, and configurations across the considered model domain. Figure 4-15 shows how BMP details are summarized by capacity, cost, and footprint for each jurisdiction ID—in the EWMPs/WMPs these outputs were typically called “recipes for compliance”.

|                          |           |                         |                 |                          |                |                 |                |                |                |              |  |  |  |  |  |  |  |
|--------------------------|-----------|-------------------------|-----------------|--------------------------|----------------|-----------------|----------------|----------------|----------------|--------------|--|--|--|--|--|--|--|
| Generated: 2020-04-09    |           | Scenario: 1             |                 | Assessment Point: DS All |                |                 |                |                |                |              |  |  |  |  |  |  |  |
| Evaluation Factor: SOSED |           | Target Reduction: 22.81 |                 |                          |                |                 |                |                |                |              |  |  |  |  |  |  |  |
| Capacity Units: ac-ft    |           | Footprint Units: sq-ft  |                 |                          |                |                 |                |                |                |              |  |  |  |  |  |  |  |
| Jurished ID              | Type      | Total                   | LID1            | LID4                     | LID10          | LID13           | REG1           | MJRBMP1        | MJRBMP2        | MJRBMP3      |  |  |  |  |  |  |  |
| 700549                   | Capacity  | 18.7071                 | 8.3189          | 0.5795                   | 0.3597         | 2.8116          | 4.2040         | 2.4334         | 0.0000         | 0.0000       |  |  |  |  |  |  |  |
| 700549                   | Footprint | 206744.6                | 154201.0        | 9178.6                   | 13045.3        | 1973.8          | 17953.8        | 10392.2        | 0.0            | 0.0          |  |  |  |  |  |  |  |
| 700549                   | Cost      | \$120,414,905.39        | \$76,329,493.07 | \$4,928,904.65           | \$4,631,084.87 | \$32,210,614.98 | \$2,375,282.49 | \$1,769,784.31 | \$0.00         | \$0.00       |  |  |  |  |  |  |  |
| 700649                   | Capacity  | 3.0455                  | 0.0037          | 0.0000                   | 0.0000         | 0.0000          | 0.0000         | 0.0000         | 3.0418         | 0.0000       |  |  |  |  |  |  |  |
| 700649                   | Footprint | 13059.1                 | 68.9            | 0.0                      | 0.0            | 0.0             | 0.0            | 0.0            | 12990.2        | 0.0          |  |  |  |  |  |  |  |
| 700649                   | Cost      | \$2,065,410.92          | \$34,124.49     | \$0.00                   | \$0.00         | \$0.00          | \$0.00         | \$0.00         | \$2,116,102.94 | \$0.00       |  |  |  |  |  |  |  |
| 700787                   | Capacity  | 0.0000                  | 0.0000          | 0.0000                   | 0.0000         | 0.0000          | 0.0000         | 0.0000         | 0.0000         | 0.0000       |  |  |  |  |  |  |  |
| 700787                   | Footprint | 0.0                     | 0.0             | 0.0                      | 0.0            | 0.0             | 0.0            | 0.0            | 0.0            | 0.0          |  |  |  |  |  |  |  |
| 700787                   | Cost      | \$0.00                  | \$0.00          | \$0.00                   | \$0.00         | \$0.00          | \$0.00         | \$0.00         | \$0.00         | \$0.00       |  |  |  |  |  |  |  |
| 700849                   | Capacity  | 0.3738                  | 0.0000          | 0.0000                   | 0.0000         | 0.0179          | 0.0000         | 0.0000         | 0.0000         | 0.3559       |  |  |  |  |  |  |  |
| 700849                   | Footprint | 1532.5                  | 0.0             | 0.0                      | 0.0            | 12.6            | 0.0            | 0.0            | 0.0            | 1519.9       |  |  |  |  |  |  |  |
| 700849                   | Cost      | \$1,250,917.59          | \$0.00          | \$0.00                   | \$0.00         | \$205,304.15    | \$0.00         | \$0.00         | \$0.00         | \$244,022.34 |  |  |  |  |  |  |  |
| 700949                   | Capacity  | 0.4853                  | 0.0080          | 0.0000                   | 0.0014         | 0.0000          | 0.0000         | 0.0000         | 0.0000         | 0.4759       |  |  |  |  |  |  |  |
| 700949                   | Footprint | 2231.0                  | 148.4           | 0.0                      | 50.1           | 0.0             | 0.0            | 0.0            | 0.0            | 2032.5       |  |  |  |  |  |  |  |
| 700949                   | Cost      | \$1,229,520.03          | \$73,439.99     | \$0.00                   | \$17,788.76    | \$0.00          | \$0.00         | \$0.00         | \$0.00         | \$326,316.19 |  |  |  |  |  |  |  |
| 700983                   | Capacity  | 0.1455                  | 0.0000          | 0.1010                   | 0.0444         | 0.0000          | 0.0000         | 0.0000         | 0.0000         | 0.0000       |  |  |  |  |  |  |  |
| 700983                   | Footprint | 3212.0                  | 0.0             | 1600.2                   | 1611.8         | 0.0             | 0.0            | 0.0            | 0.0            | 0.0          |  |  |  |  |  |  |  |
| 700983                   | Cost      | \$1,288,552.26          | \$0.00          | \$859,323.54             | \$572,185.83   | \$0.00          | \$0.00         | \$0.00         | \$0.00         | \$0.00       |  |  |  |  |  |  |  |
| 701049                   | Capacity  | 0.0000                  | 0.0000          | 0.0000                   | 0.0000         | 0.0000          | 0.0000         | 0.0000         | 0.0000         | 0.0000       |  |  |  |  |  |  |  |
| 701049                   | Footprint | 0.0                     | 0.0             | 0.0                      | 0.0            | 0.0             | 0.0            | 0.0            | 0.0            | 0.0          |  |  |  |  |  |  |  |
| 701049                   | Cost      | \$0.00                  | \$0.00          | \$0.00                   | \$0.00         | \$0.00          | \$0.00         | \$0.00         | \$0.00         | \$0.00       |  |  |  |  |  |  |  |
| 701087                   | Capacity  | 0.0000                  | 0.0000          | 0.0000                   | 0.0000         | 0.0000          | 0.0000         | 0.0000         | 0.0000         | 0.0000       |  |  |  |  |  |  |  |
| 701087                   | Footprint | 0.0                     | 0.0             | 0.0                      | 0.0            | 0.0             | 0.0            | 0.0            | 0.0            | 0.0          |  |  |  |  |  |  |  |
| 701087                   | Cost      | \$0.00                  | \$0.00          | \$0.00                   | \$0.00         | \$0.00          | \$0.00         | \$0.00         | \$0.00         | \$0.00       |  |  |  |  |  |  |  |
| 701187                   | Capacity  | 0.0000                  | 0.0000          | 0.0000                   | 0.0000         | 0.0000          | 0.0000         | 0.0000         | 0.0000         | 0.0000       |  |  |  |  |  |  |  |
| 701187                   | Footprint | 0.0                     | 0.0             | 0.0                      | 0.0            | 0.0             | 0.0            | 0.0            | 0.0            | 0.0          |  |  |  |  |  |  |  |
| 701187                   | Cost      | \$0.00                  | \$0.00          | \$0.00                   | \$0.00         | \$0.00          | \$0.00         | \$0.00         | \$0.00         | \$0.00       |  |  |  |  |  |  |  |
| Total                    | Capacity  | 22.7572                 | 8.3306          | 0.6805                   | 0.4055         | 2.8295          | 4.2040         | 2.4334         | 3.0418         | 0.8318       |  |  |  |  |  |  |  |
| Total                    | Footprint | 226779.2                | 154418.3        | 10778.8                  | 14707.2        | 1986.4          | 17953.8        | 10392.2        | 12990.2        | 3552.4       |  |  |  |  |  |  |  |
| Total                    | Cost      | \$126,249,306.20        | \$76,437,057.55 | \$5,788,228.19           | \$5,221,059.46 | \$32,415,919.13 | \$2,375,282.49 | \$1,769,784.31 | \$2,116,102.94 | \$570,338.52 |  |  |  |  |  |  |  |

Figure 4-15. Example output from the Implementation Plan Generator.

- ▼ **Instream Post-implementation Timeseries Utility.** The Instream Post-implementation Timeseries Utility provides a model architecture to link output from the SUSTAIN Optimization Framework back to the WMMS 2.0 calibrated LSPC watershed model. To assess the hydrologic and water quality impact that the selected BMPs would have on the existing condition in LSPC, this Utility generates a batch file to execute a series of SUSTAIN runs that culminate in flow and pollutant load timeseries for each jurisdiction. Timeseries are generated for the outlet from the most downstream MJR BMP. Those timeseries can be routed back into the WMMS 2.0 LSPC model,

replacing the baseline runoff (i.e., SURO) boundary conditions. This utility enables users to evaluate the routed impact of BMPs on instream hydrology and pollutant levels and compare against water quality standards.

The tools listed above and the utilities in the Optimization Framework represent a major leap forward in providing a consistent, transparent, and user-focused optimization-based modeling platform for BMP implementation planning.

## 5 SUMMARY AND CONCLUSION

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WMMS 2.0 includes numerous advancements in BMP modeling processes and cost-optimization procedures of SUSTAIN that improve model functionality, transparency, and accessibility for use as a decision support tool for BMP implementation planning and watershed management. The following provides a summary of these advancements and resulting tools:

- ▼ WMMS 2.0 includes multiple enhancements to the SUSTAIN modeling platform for improved efficiency (e.g., upgrade to 64-bit environment), BMP modeling functionality, (e.g., underdrain effluent concentration, BMP diversion, BMP bypass constituent concentration, representation of PET, BMP discharge rate), and procedures for BMP optimization (e.g., 2-tiered framework). Building upon SUSTAIN version 1.2 release by EPA in 2014, this new version of SUSTAIN is tailored to support BMP implementation planning at the watershed scale, and can better accommodate the typical BMPs included within E/WMPs in the Los Angeles region.
- ▼ WMMS 2.0 capitalizes on lessons learned from E/WMPs developed in the Los Angeles region and BMP implementation throughout California and provides default BMP modeling parameters tailored for southern California that reflect the state-of-the-practice in considerations for BMP design, performance, and costs. These default parameters and associated assumptions are documented in this report. These default parameters provide model users with a firm foundation to initiate the use of WMMS 2.0 for BMP modeling, however, parameters can be further modified based on user-preferences or other considerations. This results in a system that is adaptable and is capable of continuous updates over time to reflect improved understanding of BMP design and implementation that can result from continued research in the region.
- ▼ The WMMS 2.0 web-based modeling platform includes a number of utilities to enable visualization of model output and to facilitate the process for BMP cost-optimization. The resulting system provides a convenient framework to support Watershed Management Groups with the development and updates of RAAs using a consistent approach. The utilities provide linkages to WRAMPS in order to streamline processes that considered implemented BMPs, standardized processes for entering information on planned future BMPs, and tools to support the 2-tiered optimization procedures and methods to visualize and access results that serve as decision-support for watershed management.

## 6 REFERENCES

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- Ackerman, D., K. Schiff, S. Weisberg. 2005. Evaluating HSPF in an Arid, Urbanized Watershed. *Journal of the American Water Resources Association*, pp 477-486.
- Ackerman, D. and S. Weisberg. 2006. Evaluating HSPF Runoff and Water Quality Predictions at Multiple Time and Spatial Scales. *Southern California Coastal Water Research Project 2005-06 Biennial Report*. Southern California Coastal Water Research Project, Westminster, CA.



- Ackerman, D. and E. Stein. 2008. Evaluating the Effectiveness of Best Management Practices Using Dynamic Modeling. *Journal of Environmental Engineering*, pp 628–639.
- Akan, A. 1993. *Urban Stormwater Hydrology: A guide to Engineering Calculations*. CRC Press: Washington D.C.
- Ballona Creek WMG (Watershed Management Group). 2016. *Enhanced Watershed Management Program for the Ballona Creek Watershed*. Prepared by Black and Veatch, CDM Smith, CH2M, Larry Walker and Associates, Paradigm Environmental, and Tetra Tech for the Ballona Creek Watershed Management Group.
- Boulanger, B. and N. Nikolaidis. 2003. Modeling Framework for Managing Copper in Urban Watersheds. *Journal of American Water Resources Association (JAWRA)*, 39(2) pp 337–345.
- Brown, J. and B. Peake. 2006. Sources of Heavy Metals and Polycyclic Aromatic Hydrocarbons in Urban Stormwater Runoff. *Science of The Total Environment*, 359(1–3) pp 145–55.
- CASQA (California Stormwater Quality Association). 2003. *California Stormwater BMP Handbook New Development and Redevelopment*.
- CASQA-LIDI (California Stormwater Quality Association-Low Impact Development Initiative). 2017a. Low Impact Development Stormwater Management Standard Details.
- CASQA-LIDI (California Stormwater Quality Association-Low Impact Development Initiative). 2017b. Attachment A: Bioretention Soil Specification.
- City of Alexandria. 2015. Long Term Control Plan Update Basis for Cost Opinions. Prepared by Greeley and Hansen for the City of Alexandria Department of Transportation and Environmental Quality.
- City of Alhambra. 2018. Regional Stormwater Capture Project at Almansor Park Project Concept Sheet. Prepared by Paradigm Environmental for the City of Alhambra.
- City of LA. 2006a. *La Cienega/Fairfax Powerline Easement Stormwater Best Management Practices (Baldwin Hills to Ballona Creek Infiltration) Project Concept Report*. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division. Santa Monica Bay Beaches Bacterial TMDL Implementation Plan.
- City of LA. 2006b. *Mar Vista Recreation Center Stormwater Best Management Practices Project Concept Report*. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division. Santa Monica Bay Beaches Bacterial TMDL Implementation Plan.
- City of LA. 2006c. *Westchester Recreation Center Stormwater Best Management Practices Project Concept Report*. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division. Santa Monica Bay Beaches Bacterial TMDL Implementation Plan.
- City of LA. 2006d. *South Los Angeles Wetland Park Proposition O Project Draft Project Concept Report*. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division.
- City of LA. 2006e. *Grand Avenue Stormwater Best Management Practices Project Draft Concept Report*. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division. Santa Monica Bay Beaches Bacterial TMDL Implementation Plan.
- City of LA. 2006f. *Imperial Highway Stormwater Best Management Practices Project Draft Concept Report*. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division. Santa Monica Bay Beaches Bacterial TMDL Implementation Plan.

- City of LA. 2009. Green Streets & Green Alleys Design Guidelines Standards. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division. Rainwater Harvesting Program.
- City of LA. 2011. *Rancho Cienega Sports Complex Regional BMP Project Final Project Concept Report*. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division.
- City of LA. 2012. *Del Rey Lagoon Water Quality Improvement Project Draft Concept Report*. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division.
- City of LA. 2013. *Final Project Concept Report Albion Riverside Park Project*. City of Los Angeles Department of Public Works Bureau of Sanitation Watershed Protection Division.
- City of LA. 2014a. *Mendeville Canyon BMP Project Concept Report*. City of Los Angeles. Santa Monica Bay Best Management Practices Strategy.
- City of LA. 2014b. *Santa Ynez Canyon BMP Project Concept Report*. City of Los Angeles. Santa Monica Bay Best Management Practices Strategy.
- City of LA. 2014c. *Complete Streets Manual: Chapter 9 of the City of Los Angeles Mobility Plan*. City of Los Angeles Department of City Planning.
- City of LA. 2015. Riviera Country Club Stormwater BMP Project, TOS SN-22 Concept Summary. City of Los Angeles.
- City of LA. 2016a. Venice Boulevard Neighborhood Green Street Regional Project: Penmar Street – North of Venice Boulevard. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division. Ballona Creek, Santa Monica Bay, & Marina del Rey TMDL & EWMP Implementation Section.
- City of LA. 2016b. *South Occidental Boulevard Stormwater BMP Concept Report*. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division. Ballona Creek, Santa Monica Bay, & Marina del Rey TMDL & EWMP Implementation Section.
- City of LA. 2017a. South Hoover Forebay Green Street BMP. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division. Ballona Creek, Santa Monica Bay, & Marina del Rey TMDL & EWMP Implementation Section.
- City of LA. 2017b. West Manchester Avenue Green Street BMP. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division. Ballona Creek, Santa Monica Bay, & Marina del Rey TMDL & EWMP Implementation Section.
- City of LA. 2017c. Pico-olive Stormwater BMP. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division. Ballona Creek, Santa Monica Bay, & Marina del Rey TMDL & EWMP Implementation Section.
- City of LA. 2017d. *Queen Anne Recreational Center Signature Project Pre-Design Report*. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division. Ballona Creek, Santa Monica Bay, & Marina del Rey TMDL & EWMP Implementation Section.
- City of LA. 2017f. *Ballona Creek Venice Boulevard Green Street Concept Report*. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division. Ballona Creek, Santa Monica Bay, & Marina del Rey TMDL & EWMP Implementation Section.

- City of LA. 2017g. *Cochran Avenue Green Street Concept Report*. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division. Ballona Creek, Santa Monica Bay, & Marina del Rey TMDL & EWMP Implementation Section.
- City of LA. 2017h. *Magnolia Avenue Rain Garden Parklette and Green Street Concept Report*. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division. Ballona Creek, Santa Monica Bay, & Marina del Rey TMDL & EWMP Implementation Section.
- City of LA. 2017i. *Kittyhawk Green Walkway Revised Project Proposal*. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division.
- City of LA. 2017j. *Palms Boulevard Median BMP Concept Report*. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division. Ballona Creek, Santa Monica Bay, & Marina del Rey TMDL & EWMP Implementation Section.
- City of LA. 2017k. *Rose Avenue Green Street Concept Report*. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division. Ballona Creek, Santa Monica Bay, & Marina del Rey TMDL & EWMP Implementation Section.
- City of LA. 2017l. *Vermont Avenue and Stormwater Capture Green Street Concept Report*. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division.
- City of LA. 2018. *Permeable Pavers at Parkway - Standard Plan: S-489-0*. City of Los Angeles Department of Public Works Bureau of Engineering.
- City of LA. 2019. *Proposition O – Clean Water Bond Program: February 2019 Monthly Report*. City of Los Angeles Bureau of Engineering, Bond Division.
- City of LA. ND (No Date). *North Pico-Union Neighborhood Stormwater Strategy*. City of Los Angeles, Department of Public Works, Bureau of Sanitation, Watershed Protection Division. Ballona Creek, Santa Monica Bay, & Marina del Rey TMDL & EWMP Implementation Section.
- City of San Marcos. 2006. *Unit Prices for Bond Purposes*. City of San Marcos Engineering Division.
- CSM (Colorado School of Mines) and EPA (Environment Protection Agency). 2019. *An Integrated Decision Support Tool (I-DST) for Life-Cycle Cost Assessment and Optimization of Green, Grey, and Hybrid Stormwater Infrastructure*.
- Collins, K., W. Hunt, J. Hathaway. 2010. Side-by-side Comparison of Nitrogen Species Removal for Four Types of Permeable Pavement and Standard Asphalt in Eastern North Carolina. *Journal of Hydrologic Engineering*, pp 512–521.
- County of San Diego. 2017. *Land Development Division Unit Price List*. County of San Diego Department of Public Works.
- Crawford, N., S. Burges. 1967. History of The Stanford Watershed Model. *Water Resources Impact*, 6(2).
- Davis, A., M. Shokouhian, M. Ni. 2001. Loading Estimates of Lead, Copper, Cadmium, and Zinc in Urban Runoff from Specific Sources. *Chemosphere*, 44(5) pp 997–1009.
- Davis, A., M. Shokouhian, H. Sharma, C. Mina, D. Winogradoff. 2003. Water Quality Improvement Through Bioretention: Lead, Copper, and Zinc Removal. *Water Environment Research*, 75(1): 73–82.
- Davis, A. and R. McCuen. 2005. *Stormwater Management for Smart Growth*. Springer: New York.



- Deb, K. A. Pratap, S. Agarwal, T. Meyarivan. 2002. A Fast and Elitist Multi-objective Genetic Algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation*, 6(2) pp 182-197
- DC WMG (Dominguez Channel Watershed Management Group). 2016. *Enhanced Watershed Management Program for the Dominguez Channel Watershed Management Area Group*.
- ESGV WMG (East San Gabriel Valley Watershed Management Group). 2014. *Watershed Management Program Plan for the East San Gabriel Valley Watershed Management Group*. Prepared for the East San Gabriel Valley Watershed Management Group by MWH, in conjunction with Larry Walker Associates, Paradigm Environmental, and Tetra Tech.
- E-commerce Retailer. 2019. Bulk Data download from [www.plastic-mart.com](http://www.plastic-mart.com). Extracted by Paradigm Environmental in March 2019.
- Frere, M, C. Onstad, H. Holtan. 1975. *ACTMO, An Agricultural Chemical Transport Model*. United States Department of Agriculture, Agricultural Research Service.
- Gallo, E., T. Hogue, C. Bell, K. Spahr, E. Grubert, J. Stokes-Draut. 2018. Application of a New integrated Decision Support Tool (i-DST) for Urban Water Infrastructure: Analyzing Water Quality Compliance Pathways in Ballona Creek Watershed in Los Angeles, California; American Geophysical Union, Fall Meeting 2018.
- Gao, J., R. Wang, J. Huang, M. Liu. 2015. Application of BMP to Urban Runoff Control Using SUSTAIN Model: Case Study in an Industrial Area. *Ecological Modelling*, 318(12) pp 177–183.
- Gold, M., T. Hogue, S. Pincetl, K. Mika, K. Radavich. 2015. *LA Sustainable Water Project: Ballona Creek Watershed*. University of California Los Angeles: Sustainable LA Grand Challenge.
- Ha, H. and M. Stenstrom. 2003. Identification of Land Use with Water Quality Data in Stormwater Using a Neural Network. *Water Research*, 37(17) pp 4222–4230.
- Holtan, H and N. Creitz. 1967. Influence of Soils, Vegetation and Geomorphology on Elements of the Flood Hydrograph. Symposium on Floods and their Computation; Leningrad, Russia.
- Homer, C., J. Dewitz, L. Yang, S. Jin, P. Danielson, G. Xian, J. Coulston, N. Herold, J. Wickham, K. Megown. 2004. Completion of the 2011 National Land Cover Database for the conterminous United States-Representing a Decade of Land Cover Change Information. *Photogrammetric Engineering and Remote Sensing*, 81(5) pp 345-354.
- Houle, J, R. Roseen, T. Ballesterio, T. Puls, J. Sherrard. 2013. Comparison of Maintenance Cost, Labor Demands, and System Performance for LID and Conventional Stormwater Management. *Journal of Environmental Engineering*, 139 pp 932–938.
- Innovyze. 2015. InfoSWMM Product Brochure. Accessed April 2020 at <[link](#)>.
- LAC (Los Angeles County) GIS Data Portal. City Boundaries and Annexations/Detachments. Accessed February 2019 at <[link](#)>.
- LACDPW (Los Angeles County Department of Public Works). 2010a. *Los Angeles County Watershed Model Configuration and Calibration—Part I: Hydrology*. Los Angeles County Department of Public Works, Watershed Management Division.
- LACDPW (Los Angeles County Department of Public Works). 2010b. *Los Angeles County Watershed Model Configuration and Calibration—Part II: Water Quality*. Los Angeles County Department of Public Works, Watershed Management Division.
- LACDPW (Los Angeles County Department of Public Works). 2011a. *Phase III Report: Development of the Framework for Watershed-Scale Optimization Modeling*. Los Angeles County Department of Public Works, Watershed Management Division.

- LACDPW (Los Angeles County Department of Public Works). 2011b. *Water Conservation Distribution System*. Los Angeles County Department of Public Works, Water Resources Division, Operations Section Field Districts. Revised September 1, 2011.
- LACDPW (Los Angeles County Department of Public Works). 2014. *Low Impact Development Standards Manual*. Los Angeles County Department of Public Works, Watershed Management Division, Alhambra, CA.
- LACDPW (Los Angeles County Department of Public Works). 2018a. Plan Sheets for Ladera Park Stormwater Improvements. Los Angeles County Department of Public Works, Alhambra, CA.
- LACDPW (Los Angeles County Department of Public Works). 2018b. Price Bid History for Ladera Park Stormwater Improvements. Los Angeles County Department of Public Works, Alhambra, CA.
- LACDPW (Los Angeles County Department of Public Works). 2018c. Plan Sheets for Gates Canyon Stormwater Improvements Project. Los Angeles County Department of Public Works, Alhambra, CA.
- LACDPW (Los Angeles County Department of Public Works). 2018d. Price Bid History for Gates Canyon Stormwater Improvements Project. Los Angeles County Department of Public Works, Alhambra, CA.
- LACDPW (Los Angeles County Department of Public Works). 2019a. Personal Communication with TJ Moon, Los Angeles County Department of Public Works, Alhambra, CA.
- LACDPW (Los Angeles County Department of Public Works). 2019b. Watershed Reporting and Adaptive Management & Planning System (WRAMPS) – Bulk Data download. Extracted by Paradigm Environmental in March 2019. Los Angeles County Department of Public Works, Alhambra, CA.
- LACDPW (Los Angeles County Department of Public Works). 2019c. *Adventure Park 30% Design Cost Estimate*. Los Angeles County Department of Public Works, Alhambra, CA.
- LARWQCB (Los Angeles Regional Water Quality Control Board). 2012. *Waste Discharge Requirements for Municipal Separate Storm Sewer System (MS4) Discharges Within the Coastal Watersheds of Los Angeles County, Except Those Discharges Originating From The City of Long Beach MS4*. Order No. R4-2012-0175. NPDES Permit No. CAS004001.
- LARWQCB (Los Angeles Regional Water Quality Control Board). 2018. Watershed Management Programs. Accessed January 2020 at <[link](#)>.
- LARWQCB (Los Angeles Regional Water Quality Control Board) and USEPA (U.S. Environmental Protection Agency). 2005a. *Total Maximum Daily Load for Metals in Ballona Creek*. California Regional Water Quality Control Board Los Angeles Region and U.S. Environmental Protection Agency Region 9. Los Angeles, CA.
- LARWQCB (Los Angeles Regional Water Quality Control Board) and USEPA (U.S. Environmental Protection Agency). 2005b. *Total Maximum Daily Loads for Metals Los Angeles River and Tributaries*. U.S. Environmental Protection Agency, Region 9 and California Regional Water Quality Control Board, Los Angeles Region.
- LARWQCB (Los Angeles Regional Water Quality Control Board) and USEPA (U.S. Environmental Protection Agency). 2006. *Total Maximum Daily Loads for Metals and Selenium, San Gabriel River and Tributaries*. California Regional Water Quality Control Board, Los Angeles Region and U.S. Environmental Protection Agency, Region 9. Los Angeles, CA.

- LARWQCB (Los Angeles Regional Water Quality Control Board) and USEPA (U.S. Environmental Protection Agency). 2011. *Dominguez Channel and Greater Los Angeles and Long Beach Harbor Waters Toxics Total Maximum Daily Loads*. California Regional Water Quality Control Board Los Angeles Region and U.S. Environmental Protection Agency Region 9. Los Angeles, CA.
- Lai, F., C. Fan, L. Shoemaker, R. Field. 2003. Development of a Decision-support Framework for Placement of BMPs in Urban Watersheds. Water Environment Federation Specialty Conference TMDL. Nov. 15-19, 2003; Chicago, IL.
- Lai, F., L. Shoemaker, J. Riverson. 2004. Framework for Placement of BMP/LID in Urban Watersheds. LID Conference: Putting the LID on Stormwater Management. Sept. 21-23, 2004; College Park, MD.
- Lai, F., L. Shoemaker, J. Riverson. 2005. Framework Design for BMP Placement in Urban Watersheds. Proceedings of the ASCE EWRI World Water and Environmental Congress. May 15-19, 2005; Anchorage, AK.
- Lai, F., J. Zhen, J. Riverson, L. Shoemaker. 2006. SUSTAIN – An Evaluation and Cost Optimization Tool for Placement of BMPs. Proceedings of the ASCE EWRI World Water and Environmental Congress. May 21-25, 2006; Omaha, NE.
- Lai, F., T. Dai, J. Zhen, J. Riverson, K. Alvi, L. Shoemaker. 2007. SUSTAIN - An EPA BMP Process And Placement Tool For Urban Watersheds. Proceedings WEF 2007 TMDL Specialty Conference. June 24-27, 2007; Bellevue, WA.
- LCC WMP (Los Cerritos Channel Watershed Management Group). 2015. *Los Cerritos Channel Watershed Management Program*. Prepared for the Los Cerritos Channel Watershed Management Group by Richard Watson & Associates, Inc.
- LLAR WMP (Lower Los Angeles River Watershed Management Group). 2015. *Lower Los Angeles River Watershed Management Program*. Prepared for the Lower Los Angeles River Watershed Management Group by John L. Hunter & Associates, Inc.
- LSGR WMP (Lower San Gabriel River Watershed Management Group). 2015. *Lower San Gabriel River Watershed Management Program*. Prepared for the Lower San Gabriel River Watershed Management Group by John L. Hunter & Associates, Inc.
- MAG (Maricopa Association of Governments). 2001. Technical Memorandum No. 1 Infrastructure Development Costs.
- Malibu Creek WMG (Malibu Creek Watershed Management Group). 2016. *Enhanced Watershed Management Program for the Malibu Creek Watershed*. Prepared by the City of Calabasas, City of Agoura Hills, City of Westlake Village, City of Hidden Hills, and County of Los Angeles, and Los Angeles County Flood Control District.
- Mandelker, D. 1989. Controlling Nonpoint Source Water Pollution. *Chicago-Kent Law Review*, 65(2) pp 479–502.
- McCuen, R. 2003. Smart Growth: Hydrologic Perspective. *Journal of Professional Issues in Engineering and Education Practice*, 129 pp 151–154.
- Mika, K., T. Hogue, S. Pincetl, E. Gallo, M. Gold. 2017a. *LA Sustainable Water Project: Dominguez Channel & Machado Lake Watersheds*. University of California Los Angeles: Sustainable LA Grand Challenge.
- Mika, K., E. Gallo, L. Read, R. Edgley, T. Hogue, S. Pincetl, M. Gold. 2017b. *LA Sustainable Water Project: Los Angeles River Watershed*. University of California Los Angeles: Sustainable LA Grand Challenge.



- Musgrave, G. 1955. How Much of the Rain Enters the Soil? Water Yearbook of Agriculture. United States Department of Agriculture: Washington, DC.
- Newell, J, M. Seymour, T. Yee, J. Renteria, T. Longcore, J. Wolch, A. Shishkovsky. 2013. Green Alley Programs: Planning for a Sustainable Urban Infrastructure? *Journal of Cities*, 31 pp 144–155.
- Orange County. 2011. *Technical Guidance Document for the Preparation of Conceptual/Preliminary and/or Project Water Quality Management Plans*. Orange County Department of Public Works.
- Prince George's County. 2001. *Low-Impact Development Management Practices Evaluation Computer Module, User's Guide*. Prepared by Tetra Tech, Inc., Fairfax, VA.
- PWD (Philadelphia Water Department). 2009. *Combined Sewer Overflow Control Alternatives Costing Tool Reference Manual: Supplemental Documentation Volume 3 Basis of Cost Opinions*.
- Rawls, W., D. Brakensiek. 1982. Estimating Soil Water Retention from Soil Properties. *Journal of Irrigation and Drainage*, 108(2) pp 166–171.
- Read, K., T. Hogue, R. Edgley, K. Mika, M. Gold. 2019. Evaluating the Impacts of Stormwater Management on Streamflow Regimes in the Los Angeles River. *Journal of Water Resources and Planning Management*, 145(10).
- Riverside County. 2018. *Santa Margarita River Watershed Region Design Handbook for Low Impact Development Best Management Practices*. Riverside County Flood Control and Water Conservation District.
- Riverson, J., J. Zhen, L. Shoemaker, F. Lai. 2004. Design of a Decision Support System for Selection and Placement of BMPs in Urban Watersheds. World Water and Environmental Resources Congress. June 27-July 1, 2004; Salt Lake City, Utah.
- Rossman, L. 2008. *Stormwater Management Model User's Manual Version 5.0*. EPA/600/RM05/040. USEPA National Risk Management Research Laboratory, Office of Research and Development. Cincinnati, OH.
- SCVURPPP (Santa Clara Valley Urban Runoff Pollution Prevention Program). 2016. *Stormwater Handbook*. Sunnyvale, CA.
- Stein, E., L. Tiefenthaler, K. Schiff. 2007. *Sources, Patterns and Mechanisms of Storm Water Pollutant Loading from Watersheds and Land Uses of the Greater Los Angeles area, California, USA*. Technical Report 510. Southern California Coastal Water Research Project.
- StormTrap. 2019. Doubletrap - Design Criteria Sample Drawing.
- Taguchi, V., P. Weiss, J. Gulliver, M. Klein, R. Hozalski, L. Baker, J. Finlay, B. Keeler, J. Nieber. 2020. It Is Not Easy Being Green: Recognizing Unintended Consequences of Green Stormwater Infrastructure. *Water*, 12(2) pp 522.
- Tariq, A., R. Lempert, L. Riverson, M. Schwartz, N. Berg. 2017. A Climate Stress Test of Los Angeles' Water Quality Plans. *Climatic Change*, 144 pp 625–639.
- ULAR WMG (Upper Los Angeles River Watershed Management Group). 2016. *Enhanced Watershed Management Program for the Upper Los Angeles River Watershed*. Prepared by Black and Veatch, CH2M, Paradigm Environmental, CDM Smith, Larry Walker and Associates, and Tetra Tech for the Upper Los Angeles River Watershed Management Group.
- US Census Bureau. 2018. Urban Areas Facts.

- USBR (U.S. Department of Interior, Bureau of Reclamation) and LACDPW (Los Angeles County Department of Public Works). 2016. *Los Angeles Basin Study*. Prepared by the U.S. Department of Interior, Bureau of Reclamation, Lower Colorado Region, Southern California Area Office, Los Angeles County Department of Public Works, and Los Angeles County Flood Control District.
- USCR WMG (Upper Santa Clara River Watershed Management Group). 2016. *Upper Santa Clara River Watershed Management Group Enhanced Watershed Management Program*. Prepared for the Upper Santa Clara River Watershed Management Group by Larry Walker Associates, Paradigm Environmental, and Tetra Tech.
- USEPA (United States Environmental Protection Agency). 1997. *Compendium of Tools for Watershed Assessment and TMDL Development*. EPA 841-B-97-006. Office of Water. Washington, D.C.
- USEPA (United States Environmental Protection Agency). 1999. *Preliminary Data Summary of Urban Storm Water Best Management Practices*. EPA-821-R-99-012. Office of Water. Washington, D.C.
- USEPA (United States Environmental Protection Agency). 2008. *Handbook for Developing Watershed Plans to Restore and Protect Our Waters*. EPA 841-B-08-002. Office of Water, Washington, D.C.
- USEPA (United States Environmental Protection Agency). 2009. *SUSTAIN - A Framework for Placement of Best Management Practices in Urban Watersheds to Protect Water Quality*. Office of Research and Development. National Risk Management Research Laboratory, Water Supply and Water Resources Division.
- USEPA (United States Environmental Protection Agency). 2014. *Tool for Selecting Best Management Practices to Protect Water Quality*.
- USEPA (United States Environmental Protection Agency). 2015. *Clean Water State Revolving Fund Green Project Reserve Report*.
- USEPA (U.S. Environmental Protection Agency). 2017. *Developing Reasonable Assurance: A Guide to Performing Model-Based Analysis to Support Municipal Stormwater Program Planning*. U.S. Environmental Protection Agency Region 9, San Francisco, CA.
- USGR EWMP Group (Upper San Gabriel River Enhanced Watershed Management Program Group). 2016. *Enhanced Watershed Management Program Plan for the Upper San Gabriel River Watershed*. Prepared for the Upper San Gabriel River Enhanced Watershed Management Program Group by MWH, in conjunction with Larry Walker Associates, Paradigm Environmental, and Tetra Tech. January 2016.
- Vassilios, T. and R. Hamid. 1997. Modeling and Management of Urban Stormwater Runoff Quality: A Review. *Water Resources Management*, 11(2) pp 136–164.
- Vendor Communication. 2019a. Bio Clean: Forterra April 2019.
- Vendor Communication. 2019b. Contech Engineered Solutions April 2019.
- Vendor Communication. 2019c. StormTrap April 2019.
- Vendor Communication. 2019d. Torrent Resources April 2019.
- Weiss, P., J. Gulliver, A. Erickson. 2005. *The Cost and Effectiveness of Stormwater Management Practices*. Report 2005-23. Minnesota Department of Transportation.
- WERF (Water Environment and Reuse Foundation). 2017. *International Stormwater BMP Database, 2016 Summary Statistics, Final Report*. Prepared by Wright Water Engineers and Geosyntec Consultants.

- Wolfand, J., C. Bell, A. Boehm, T. Hogue, R. Luthy. 2018. Multiple Pathways to Bacterial Load Reduction by Stormwater Best Management Practices: Trade-offs in Performance, Volume, and Treated Area. *Environment Science & Technology*, 52(11) pp 6370–6379.
- Wong, K., E. Strecker, M. Stenstrom. 1997. A Geographic Information System to Estimate Stormwater Pollutant Mass Loadings. *Journal of Environmental Engineering*, 123(8) pp 737–745.
- Wu, J., P. Kauhanen, J. Hunt, D. Senn, T. Hale, L. McKee. 2019. Optimal Selection and Placement of Green Infrastructure in Urban Watersheds for PCB Control. *Journal of Sustainable Water in the Built Environment* 5(2). San Francisco Estuary Institute Contribution No. 729.
- Yang, L., S. Jin, P. Danielson, C. Homer, L. Gass, S. Bender, A. Case, C. Costello, J. Dewitz, J. Fry, M. Funk, B. Granneman, G. Liknes, M. Rigge, G. Xian. 2018. A New Generation of the United States National Land Cover Database—Requirements, Research Priorities, Design, and Implementation Strategies. *International Society for Photogrammetry and Remote Sensing Journal of Photogrammetry and Remote Sensing*, 146 pp 108–123.
- Zou, R., J. Riverson, Y. Liu, R. Murphy, S. Young. 2015. Enhanced Nonlinearity Interval Mapping Scheme for High-performance Simulation-optimization of Watershed-scale BMP Placement. *Water Resources Research*, 51.